
Elements of linear algebra and geometry

Vector spaces, matrices, determinants, affine and euclidean geometry

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Introduction

The aim of this module is to review the fundamental concepts and methods used in linear algebra and geometry. These concepts and methods will be necessary in the study and understanding of the following modules.

Among the conceptual foundations that are reviewed in this module are those such as vector space and subspace, linear combination, linear independence, matrices, determinants, equations of lines and planes in the three-dimensional space and some concepts of metric geometry (scalar product, orthonormality, angles and distances).

Those concepts are used in different situations: programming, as an example, uses the terminology of unidimensional arrays for naming vectors, and bidimensional arrays when talking about matrices. The equations of lines and planes in 2D and 3D, as well as the matrices themselves, have a relevant role in the fields of computer graphics. In the field of networks for telecommunications, matrix theory is used as a foundation for the theory of errors detection and correction (parity controls, linear codes, etc). It is used as well in graph theory, cryptography, etc.

The module is presented from a purely practical approach, so it includes examples that illustrate the concepts introduced, as well as outputs of different mathematical programs that can be used to streamline or check the calculations.

Objectives

The general objective of this module is to review the basic concepts and methods of linear algebra and geometry.

In particular, the teaching objectives that are intended to be achieved within this module are as follows:

1. Review the concepts associated with vector space: linear independence, generating system, base, dimension, etc.
2. Review the basic theory about matrices and determinants (operations with matrices, inverse matrix calculus, rank of a matrix, calculus and properties of determinants, etc.).
3. Review the equations of lines in 2D and 3D, as well as the equations of the planes in 3D.
4. Review and extend the concepts of dot product and vector orthogonality in $\mathbb{R}^n$.
5. Discover how math software in general can be of use to: (a) experiment with the main concepts of this topic, and (b) automate calculations and operations.
6. Explore some of the many applications of matrix theory in areas such as computing and telecommunications.

1. Introductory example

Consider a local area network (LAN) made up of several computer servers and several dozen client computers. On each day, the LAN's state (depending on the status of the servers) can be any of the following:

- State $A \rightarrow$ the LAN is working properly as no server is down.
- State $B \rightarrow$ the LAN works with problems, since some servers are down (the rest of the servers must replace it).
- Status $C \rightarrow$ the LAN is down, as all the servers are down.

From the observations of the last six months, the following table or matrix has been obtained. It shows the probabilities that the network changes from a certain state X to another state Y any day (supposing that conditions, management, use and maintenance of the LAN remain constant and will remain so for at least another six months):

Table 1. Probabilities that a red LAN will go from state X to state Y

Tomorrow (day $n + 1$)	Today (day n)		
	State A	State B	State C
State A	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
State B	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$
State C	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$

In other words: if the LAN is in state A today, the probability that tomorrow will be in state C is $1/8$; if today it is in state C , the probability that tomorrow changes to state B is $1/2$, etc.

The LAN administrator tells us that the current status of the LAN can be B or C with equal probability. That is: $P(A) = 0$, $P(B) = P(C) = 1/2$.

Introductory example

Answer the following questions:

- What is the probability that the LAN is working properly tomorrow?
- And in two days?
- And in a week?

The questions posed in this example are reasoned and answered at the end of the module.

2. Vector spaces

The concept of vector usually appears in various mathematical, physical, economic or engineering contexts. In the following sections a formal definition of vector and real vector space is given, but first we are going to review the geometric descriptions of the vector spaces of $\mathbb{R}^n$.

2.1. Vectors in $\mathbb{R}^n$

Vectors of $\mathbb{R}^2$

Given a rectangular coordinate system in the plane, we know that a certain point P is defined by an ordered pair $(p_1, p_2) \in \mathbb{R}^2$. Therefore, we can consider $\mathbb{R}^2$ as the set of all the points in the plane.

Given two points $P : (p_1, p_2)$ and $Q : (q_1, q_2)$ in $\mathbb{R}^2$, the vector $\mathbf{v} = \overrightarrow{PQ}$ is defined as the oriented segment that has its origin in P and its end in Q . The vector $\mathbf{v} = \overrightarrow{PQ}$ has components $\mathbf{v} = \overrightarrow{PQ} = (q_1 - p_1, q_2 - p_2)$. See figure 1.

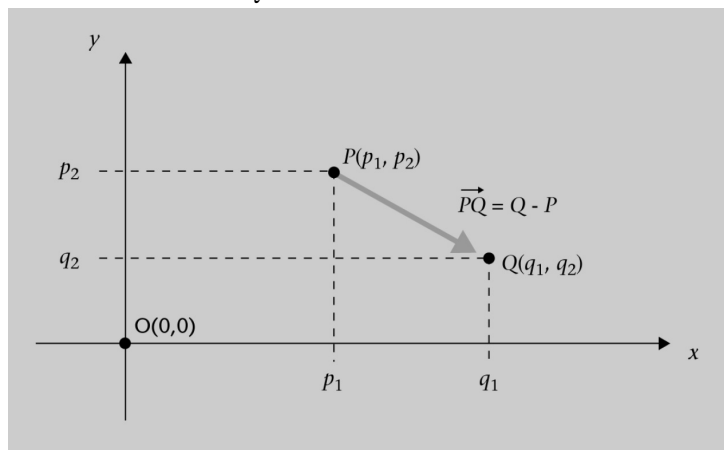
Remember

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers:
 $\mathbb{R}^n = \mathbb{R} \times \dots \times \mathbb{R}$

Equal vectors

Two vectors are considered to be equal if they have the same length and they point in the same direction, this is, they have the same components. In this case, we talk about free vectors. A free vector is determined unequivocally by its components, and in order to represent it in the plane it is enough to chose the origin or the end point.

Figure 1. Plot of the vector $\mathbf{v} = \overrightarrow{PQ}$ in $\mathbb{R}^2$



Example 1

Given the points $P : (5, 5)$ and $Q : (7, 2)$, the vector $\overrightarrow{PQ}$ has origin in P and end in Q , i.e.: $\overrightarrow{PQ} = (7, 2) - (5, 5) = (2, -3)$.

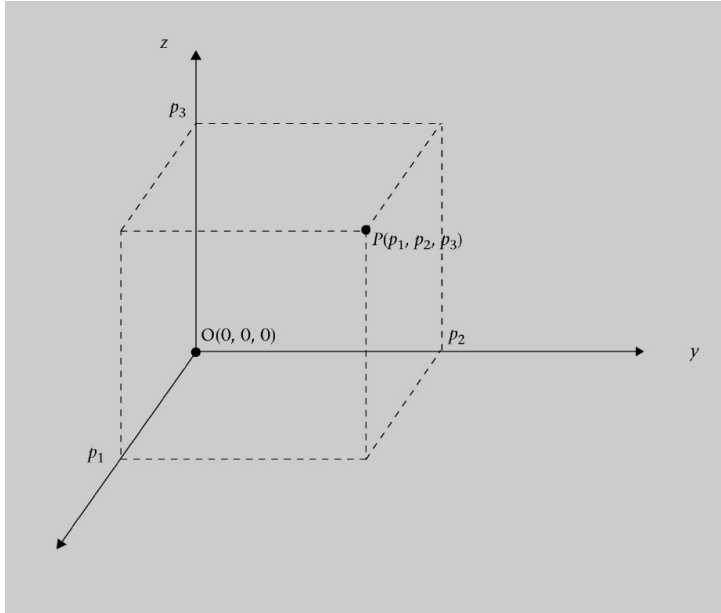
On the other hand, the vector $\overrightarrow{QP}$ has origin in Q and end in P , i.e.: $\overrightarrow{QP} = (5, 5) - (7, 2) = (-2, 3) = -\overrightarrow{PQ}$.

Their representation shows that both vectors have the same length, but opposite directions.

Vectors of $\mathbb{R}^3$

Generalizing the previous situation to $\mathbb{R}^3$, it can be considered as the set of all the points in the space represented in a system of rectangular coordinates of the space, see figure 2. The vectors of $\mathbb{R}^3$ are triplets $\mathbf{v} = (v_1, v_2, v_3)$, where a certain vector is given by its three components, and is represented as an oriented segment with origin in a point $P : (p_1, p_2, p_3)$ of $\mathbb{R}^3$ and end in a second point $Q : (q_1, q_2, q_3)$, such as $v_1 = q_1 - p_1$, $v_2 = q_2 - p_2$ and $v_3 = q_3 - p_3$.

Figure 2. Plot of the vector $\mathbf{v} = \overrightarrow{PQ}$ in $\mathbb{R}^3$



Example 2

Given the points $P : (-2, -3, 1)$ and $Q : (3, 1, 0)$, the vector $\overrightarrow{PQ}$ is the vector with origin in P and end in Q , i.e.: $\overrightarrow{PQ} = (3, 1, 0) - (-2, -3, 1) = (5, 4, -1)$.

On the other hand, the vector $\overrightarrow{QP}$ is:

$$\overrightarrow{QP} = (-2, -3, 1) - (3, 1, 0) = (-5, -4, 1) = -(5, 4, -1) = -\overrightarrow{PQ}$$

Generalizing the geometrical descriptions of the vectors of $\mathbb{R}^2$ and $\mathbb{R}^3$, we define the vectors of $\mathbb{R}^n$.

Given two points in $\mathbb{R}^n$, $P : (p_1, p_2, \dots, p_n)$ and $Q : (q_1, q_2, \dots, q_n)$, the **vector** $\mathbf{v} = \overrightarrow{PQ}$ is defined as the oriented segment that has origin in P and ends in Q . The vector $\overrightarrow{PQ}$ has components: $\mathbf{v} = \overrightarrow{PQ} = Q - P = (q_1, q_2, \dots, q_n) - (p_1, p_2, \dots, p_n) = (q_1 - p_1, q_2 - p_2, \dots, q_n - p_n)$.

Given two vectors in $\mathbb{R}^n$, $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$, and a number $k \in \mathbb{R}$, we define the following operations as follows:

Sum of vectors: $\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$

Observe that $(\mathbf{u} + \mathbf{v}) \in \mathbb{R}^n$.

Product of a scalar and a vector: $k \cdot \mathbf{u} = (k \cdot u_1, k \cdot u_2, \dots, k \cdot u_n)$

Observe that $(k \cdot \mathbf{u}) \in \mathbb{R}^n$.

Example 3

- Given two vectors $\mathbf{v} = (1, -1)$ and $\mathbf{u} = (3, -1)$, their sum is $\mathbf{v} + \mathbf{u} = (1 + 3, -1 - 1) = (4, -2)$.
- Given the vector $\mathbf{v} = (2, -3)$ and the real number $k = 3$, their scalar product is $3 \cdot \mathbf{v} = (3 \cdot 2, 3 \cdot (-3)) = (6, -9)$.

Example 4

- Given two vectors $\mathbf{v} = (1, -1, 0)$ and $\mathbf{u} = (3, -1, 1)$, their sum is $\mathbf{v} + \mathbf{u} = (1 + 3, -1 - 1, 0 + 1) = (4, -2, 1)$.
- Given the vector $\mathbf{v} = (2, -3, 1)$ and the real number $k = 3$, their scalar product is $3 \cdot \mathbf{v} = (3 \cdot 2, 3 \cdot (-3), 3 \cdot 1) = (6, -9, 3)$.

2.2. Definition of real vector space

In the previous section we have considered the set of vectors $\mathbb{R}^n$ ($n \geq 1$) and two operations have been defined on it: the sum of vectors (+) and the product of a vector by a scalar ($\cdot$). These operations meet certain properties (associative, commutative, existence of zero element, etc.), and endow the set $\mathbb{R}^n$ with a special structure called *vector space*. In this section, we will generalize this concept:

Mathematical notation

- $\mathbf{u} \in V$ reads “ $\mathbf{u}$ belongs to the set V ”.
- $\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ reads “For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ belonging to V ”.
- $\forall k, h, w \in \mathbb{R}$ reads “For all k, h, w belonging to $\mathbb{R}$ ”.
- $\exists 0 \in V$ reads “Exists 0 belonging to V ”.

Given a set V and two operations defined on it, the sum of elements of V ($+$) and the product of an element of V by a real number ($\cdot$), the combination $(V, +, \cdot)$ is called a **real vector space** if, for all $u, v, w \in V$ and $k, h \in \mathbb{R}$, the following properties hold:

Sum

- Associative: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- Commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- Existence of zero: $\exists 0 \in V$ such that $\mathbf{u} + 0 = 0 + \mathbf{u} = \mathbf{u}$.
- Existence of opposed element: $\forall \mathbf{u} \in V, \exists \mathbf{v} \in V$ such that $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u} = 0$.
Such $\mathbf{v}$ is usually denoted by $(-\mathbf{u})$.

Product

- Distributive I: $k \cdot (\mathbf{u} + \mathbf{v}) = k \cdot \mathbf{u} + k \cdot \mathbf{v}$
- Distributive II: $(k + h) \cdot \mathbf{u} = k \cdot \mathbf{u} + h \cdot \mathbf{u}$
- Associative: $k \cdot (h \cdot \mathbf{u}) = (k \cdot h) \cdot \mathbf{u}$
- Existence of identity element: $1 \cdot \mathbf{u} = \mathbf{u}$

The elements of a vector space are called **vectors**.

Example 5. Real vector spaces

- Given $n \geq 1$, $(\mathbb{R}^n, +, \cdot)$ is a real vector space. In particular, $(\mathbb{R}^2, +, \cdot)$ is the space of the vectors in the plane, and $(\mathbb{R}^3, +, \cdot)$ is the space of the vectors in the space.
- Given $n \geq 1$, $(\mathbb{R}_n[x], +, \cdot)$ is a real vector space, where $\mathbb{R}_n[x]$ is the set of polynomials of degree lesser or equal than n , $(+)$ the sum of polynomials, and $(\cdot)$ the product of a polynomial by a real number.
- Given a certain interval $I \subset \mathbb{R}$, $(C_I(x), +, \cdot)$ is a real vector space, where $C_I(x)$ is the set of continuous real-valued functions defined on I , $(+)$ the sum of functions, and $(\cdot)$ the product of a function by a real number.

Proposition: Let $(V, +, \cdot)$ be a real vector space, $\forall \mathbf{u} \in V$ and $\forall k \in \mathbb{R}$ it holds:

- $0 \cdot \mathbf{u} = 0$
- $(-1) \cdot \mathbf{u} = -\mathbf{u}$
- $k \cdot 0 = 0$
- $k \cdot \mathbf{u} = 0 \iff [k = 0 \text{ or } \mathbf{u} = 0]$

Vector subspace

Consider the real vector space $(\mathbb{R}^3, +, \cdot)$ and the nonempty subset $W = \{(x, y, z) \in \mathbb{R}^3 | z = 0\}$ (this is, W is the xy plane of $\mathbb{R}^3$). The structure $(W, +, \cdot)$ verifies that both the sum of two elements of W , as well as the product of a scalar with an

element of W , give as a result another element of W . Therefore, W is considered to be a vector subspace of $(\mathbb{R}^3, +, \cdot)$. In general, we have:

Let $(V, +, \cdot)$ be a real vector space, we say that $(W, +, \cdot)$ is a vector subspace of $(V, +, \cdot)$ if the following conditions hold:

- 1) $\emptyset \neq W \subseteq V$
- 2) $\forall \mathbf{u}, \mathbf{w} \in W, \mathbf{u} + \mathbf{w} \in W$
- 3) $\forall \mathbf{u} \in W, \forall k \in \mathbb{R}, k \cdot \mathbf{u} \in W$

In this case, we use the notation $W \leq V$.

Another example of vector subspace of $(\mathbb{R}^3, +, \cdot)$ is given by the subset $W = \{(x, y, z) \in \mathbb{R}^3 | x + y + z = 0\}$. Its graphical representation is also a plane of $\mathbb{R}^3$. (On the other hand, a plane of $\mathbb{R}^3$ which doesn't contain the origin isn't a vector subspace of $\mathbb{R}^3$, because $(0, 0, 0)$ wouldn't belong to W).

2.3. Linear combination. Generated subspace

In this section we will study one of the most common methods to get vector subspaces.

Let $(V, +, \cdot)$ be a real vector space, we say that the vector $\mathbf{v} \in V$ is a **linear combination** of the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n \in V$ if there exists $k_1, k_2, \dots, k_n$ real numbers such that

$$\mathbf{v} = k_1 \cdot \mathbf{u}_1 + k_2 \cdot \mathbf{u}_2 + \dots + k_n \cdot \mathbf{u}_n$$

Example 6

- a) In $(\mathbb{R}^2, +, \cdot)$, the vector $(1, -3)$ is a linear combination of the vectors $(1, 0)$ and $(0, 1)$ because:

$$(1, -3) = 1 \cdot (1, 0) + (-3) \cdot (0, 1).$$

- b) In $(\mathbb{R}^3, +, \cdot)$, the vector $(2, -3, 1)$ is a linear combination of the vectors $(1, 0, 0)$,

$(0, 3, 1)$ and $(0, 0, 1)$, such that:

$$(2, -3, 1) = 2 \cdot (1, 0, 0) + (-1) \cdot (0, 3, 1) + 2 \cdot (0, 0, 1).$$

It can be proven that, given a set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$, the set of all their possible linear combinations is a vector subspace.

Let $(V, +, \cdot)$ be a real vector space, and $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ a set of vectors from V . We call **vector space generated by $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$** to the vector subspace of V consisting of all the possible linear combinations of the vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$, i.e.:

$$\langle \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n \rangle = \{k_1 \cdot \mathbf{u}_1 + k_2 \cdot \mathbf{u}_2 + \dots + k_n \cdot \mathbf{u}_n \mid k_i \in \mathbb{R} \forall i = 1, 2, \dots, n\}$$

The set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is called a **generating system** of the vector space $\langle \mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n \rangle$.

Example 7

Given the vectors $(1, -1, 3)$ and $(2, -5, 6)$ of $\mathbb{R}^3$, the vector space generated by these vectors is:

$$\begin{aligned} \langle (1, -1, 3), (2, -5, 6) \rangle &= \{k \cdot (1, -1, 3) + h \cdot (2, -5, 6) \mid k, h \in \mathbb{R}\} \\ &= \{(k + 2h, -k - 5h, 3k + 6h) \mid k, h \in \mathbb{R}\} \end{aligned}$$

Example 8

The set of vectors $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0)\}$ is a generating system of $\mathbb{R}^3$, since given any $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3$, we have

$$\mathbf{u} = u_1 \cdot (1, 0, 0) + u_2 \cdot (0, 1, 0) + u_3 \cdot (0, 0, 1)$$

Therefore $\langle (1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0) \rangle = \mathbb{R}^3$.

Observe that $\langle (1, 0, 0), (0, 1, 0), (0, 0, 1) \rangle = \mathbb{R}^3$ is also true.

2.4. Linear dependence and independence. Basis and dimension of a vector space

Given a vector space and a set of vectors, we say that these are linearly dependent if one of them can be written as a linear combination of the others. Otherwise, the vectors are said to be linearly independent. Formally:

Let $(V, +, \cdot)$ be a vector space, and $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ a set of vectors of V . The vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ are **linearly dependent** if there exist any $i \in \{1, 2, \dots, n\}$ such that $\mathbf{u}_i$ can be written as a linear combination of the other vectors, i.e.:

$$\mathbf{u}_i = k_1 \cdot \mathbf{u}_1 + \dots + k_{i-1} \cdot \mathbf{u}_{i-1} + k_{i+1} \cdot \mathbf{u}_{i+1} + \dots + k_n \cdot \mathbf{u}_n$$

i.e.: the equation $k_1 \cdot \mathbf{u}_1 + k_2 \cdot \mathbf{u}_2 + \dots + k_n \cdot \mathbf{u}_n = 0$ has a non-trivial solution.

On the contrary, $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ are **linearly independent** if the previous condition doesn't hold, i.e.: the equation $k_1 \cdot \mathbf{u}_1 + k_2 \cdot \mathbf{u}_2 + \dots + k_n \cdot \mathbf{u}_n = 0$ only has the trivial solution $k_i = 0, i = 1, 2, \dots, n$.

Example 9

The vectors $(3, 3, 2), (1, 1, -1), (2, 2, 3)$ are linearly dependent, since the first vector $(3, 3, 2)$ can be written as a linear combination of the other two: taking $k = h = 1$ it holds:

$$(3, 3, 2) = k \cdot (1, 1, -1) + h \cdot (2, 2, 3) = (k + 2h, k + 2h, -k + 3h)$$

Example 10

The vectors $(1, 0)$ and $(0, 1)$ are linearly independent. Indeed, considering the equation:

$$k \cdot (1, 0) + h \cdot (0, 1) = (0, 0)$$

we have $(k, 0) + (0, h) = (k, h) = (0, 0)$, therefore $k = h = 0$.

We define the **rank of a set of vectors** as the maximum number of linearly independent vectors.

Example 11

- a) The set of vectors $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ has rank 3, because the three vectors are linearly independent.
- b) The set of vectors $\{(1, 2, 3), (0, 2, 2), (1, 4, 5)\}$ has rank 2, because there are only two linearly independent vectors (the third vector is the sum of the other two).
- c) The set of vectors $\{(1, 0, 2), (2, 0, 4), (3, 0, 6)\}$ has rank 1, because there is only a single linearly independent vector (the second and third are multiples of the first one).

Let $(V, +, \cdot)$ be a real vector space, and $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ a set of vectors from V such that:

- 1) B is a generating system of V .
- 2) B consists only of linearly independent vectors.

We call B a **basis** of V , and we say that the **dimension** of V is n ($\dim(V) = n$).

Example 12

The set $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is a generating system of $\mathbb{R}^3$ and, in addition, the vectors of B are linearly independent. Therefore, B is a basis of $\mathbb{R}^3$ and the dimension of this vector space is 3.

Another of the many possible basis of $\mathbb{R}^3$ could be, for example, $B' = \{(1, 1, 0), (0, 2, 0), (1, 0, 4)\}$.

Note

This definition makes sense because it can be proven that all the bases of V have the same number of vectors.

In $\mathbb{R}^n$, the set of n vectors

$$B = \{(1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1)\}$$

is a basis called **canonical basis**.

It is usually denoted by $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, \dots, 0)$, ..., $\mathbf{e}_n = (0, 0, \dots, 1)$.

Proposition. Let $(V, +, \cdot)$ be a real vector space of dimension n .

- 1) Every generating system of V will have at least n vectors.
- 2) Every linearly independent set of V will have at most n vectors.
- 3) n linearly independent vectors form a basis.
- 4) A generating system of V with n vectors is a basis.

Example 13

The set of vectors $\{(1, 0, -1), (2, 3, 1)\}$ cannot be a generating set of all $\mathbb{R}^3$, since the dimension of $\mathbb{R}^3$ is 3 and, therefore, any generating system of this space must be composed of 3 or more vectors.

On the other hand, the vectors $\{(1, 0, -1), (2, 3, 1), (0, 1, 1), (0, 1, -3)\}$ cannot be linearly independent, since the dimension of $\mathbb{R}^3$ is 3, and no set with more than 3 vectors will be linearly independent.

Example 14

The set of vectors $\{(1, 0, -1), (0, 3, 1), (1, 1, 0)\}$ is a basis of $\mathbb{R}^3$, because they are linearly independent.

The set of vectors $\{(1, 0, 0), (0, 0, 1), (1, 1, 0)\}$ is a basis of $\mathbb{R}^3$, because they are a generating system of $\mathbb{R}^3$ consisting of three vectors.

Components of a vector in a basis

The concept of basis of a vector space V leads to the definition of a kind of “coordinate system” of V . The key point that will allow us to define it is the following theorem of unique representation of a vector in a basis:

Theorem. Let $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be a basis of the real vector space V . For every vector $\mathbf{v} \in V$ there exists a unique set of real numbers $c_1, \dots, c_n$ such that:

$$\mathbf{v} = c_1 \cdot \mathbf{u}_1 + \dots + c_n \cdot \mathbf{u}_n$$

We call $c_1, \dots, c_n$ **components** of $\mathbf{v}$ in the basis B .

Note

Using the components of a vector in a certain basis lets us think of any real vector space of dimension n as if was $\mathbb{R}^n$ because any vector can be written as a set of n numbers.

Example 15

Consider the basis $B = \{(1, 0), (2, 1)\}$ of $\mathbb{R}^2$. The components of the vector $(-2, 3)$ in this basis are $c_1, c_2 \in \mathbb{R}$ such that $(-2, 3) = c_1 \cdot (1, 0) + c_2 \cdot (2, 1) = (c_1 + 2c_2, c_2)$. Therefore, $c_1 = -8$ and $c_2 = 3$. In conclusion, the components of $(-2, 3)$ in the basis B are -8 and 3 .

Observe that the components of $(-2, 3)$ in the canonical basis of $\mathbb{R}^2$ are precisely -2 and 3 .

Dimension of a subspace

The definition of basis of a vector space can be generalized to subspaces:

Let W be a linear subspace of a vector space V . A set of vectors $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ from W is a basis of the subspace W if and only if:

- 1) B is a generating system of W .
- 2) B is a linearly independent set.

We call **dimension of the subspace W** ($\dim(W)$) to the number of vectors of a basis.

3. Matrices

3.1. Concept of matrix

Solving many problems we use linear systems of equations, such as:

$$\begin{cases} x + 4y - z &= 4 \\ -x + 3y + 2z &= 4 \\ 2x + 2y &= 4 \end{cases}$$

A system of linear equations such as this can be represented in a more simple way, without any risk of losing information, using a table or a matrix formed by the coefficients and the independent terms:

$$\left(\begin{array}{ccc|c} 1 & 4 & -1 & 4 \\ -1 & 3 & 2 & 4 \\ 2 & 2 & 0 & 4 \end{array} \right)$$

This table or matrix could also represent other kinds of information, such as, for example, a set of 4 vectors in $\mathbb{R}^3$ written as columns:

$$\mathbf{v}_1 = (1, -1, 2), \mathbf{v}_2 = (4, 3, 2), \mathbf{v}_3 = (-1, 2, 0), \mathbf{v}_4 = (4, 4, 4)$$

Observe that, in any case, the table or matrix is composed by 12 elements distributed in 3 rows and 4 columns.

A **matrix** $\mathbf{A}$ is a table composed by $m \times n$ elements distributed along m rows and n columns:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{matrix} \leftarrow \text{row 1} \\ \leftarrow \text{row 2} \\ \\ \leftarrow \text{row } m \end{matrix}$$

$\begin{matrix} \uparrow & \uparrow & & \uparrow \\ \text{column 1} & \text{column 2} & & \text{column } n \end{matrix}$

The matrix $\mathbf{A}$ can also be represented by (a_{ij}) , where a_{ij} references the element in the row i and column j , for $1 \leq i \leq m$ and $1 \leq j \leq n$.

It is usual to say that the matrix $\mathbf{A}$ has **dimension** $m \times n$ in order to denote that it has m rows and n columns.

Example 16

The matrix $\mathbf{A} = (a_{ij}) = \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & -5 \end{pmatrix}$ has dimension 2×3 , i.e.: it consists on 2 rows and 3 columns. It also holds that $a_{23} = -5$ and $a_{12} = -1$.

We say that two matrices $\mathbf{A}$ and $\mathbf{B}$ are equal if and only if both have the same dimension and the elements in the same place in both matrices are the same.

Example 17

The matrices $\mathbf{A} = \begin{pmatrix} 3 & b & c \\ a & 1 & 8 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} d & 7 & 4 \\ 2 & e & g \end{pmatrix}$ are equal if and only if $a = 2, b = 7, c = 4, d = 3, e = 1, g = 8$.

Given a matrix $\mathbf{A}$, the **transposed matrix** $\mathbf{A}^t$ is the matrix obtained by permuting rows by columns in $\mathbf{A}$ (this is, the first row of $\mathbf{A}$ is the first column of $\mathbf{A}^t$, the second row of $\mathbf{A}$ is the second column of $\mathbf{A}^t$, etc).

Example 18

If $\mathbf{A} = \begin{pmatrix} 3 & -2 & -1 \\ 1 & 1 & 8 \end{pmatrix}$, then $\mathbf{A}^t = \begin{pmatrix} 3 & 1 \\ -2 & 1 \\ -1 & 8 \end{pmatrix}$.

Observe that:

- a) $(\mathbf{A}^t)^t = \mathbf{A}$.
- b) $(\mathbf{A}^t)^t = \mathbf{A}$. $\mathbf{A}$ has dimension 2×3 while $\mathbf{A}^t$ has dimension 3×2 .

3.2. Special matrices

Given a matrix $\mathbf{A}$ of dimension $m \times n$, we say that:

- $\mathbf{A}$ is a **row matrix** if it has only a single row (i.e., $m = 1$).
- $\mathbf{A}$ is a **column matrix** if it has only a single column (i.e., $n = 1$).
- $\mathbf{A}$ is the **null matrix** if all its elements are 0.
- $\mathbf{A}$ is a **square matrix of order n** if it has the same number of rows than columns (i.e., $m = n$). In this case, we call **main diagonal** of the matrix to the set consisting of all the elements a_{ii} :

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \quad \leftarrow \text{main diagonal}$$

- $\mathbf{A}$ is a **symmetric square matrix** if $\mathbf{A} = \mathbf{A}^t$ (i.e., $a_{ij} = a_{ji} \forall 1 \leq i, j \leq n$).
- $\mathbf{A}$ is a **diagonal square matrix** if all the elements outside of the main diagonal are 0 (i.e., $a_{ij} = 0 \forall i \neq j$).
- $\mathbf{A}$ is the **identity matrix of order n** if it is a diagonal matrix with only 1 in the main diagonal (i.e., $a_{ij} = 0 \forall i \neq j$ and $a_{ii} = 1 \forall 1 \leq i \leq n$).
- $\mathbf{A}$ is a **upper triangular square matrix** if all the elements under the main diagonal are 0 (i.e., $a_{ij} = 0 \forall i > j$).
- $\mathbf{A}$ is a **lower triangular square matrix** if all the elements over the main diagonal are 0 (i.e., $a_{ij} = 0 \forall i < j$).

Example 19

- a) Row matrices: $\begin{pmatrix} -2 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 4 & -1 \end{pmatrix}$
- b) Column matrices: $\begin{pmatrix} 1 \\ -8 \end{pmatrix}$ $\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$
- c) Null matrices: $0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ $0_{3 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- d) Square matrices: $\begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 5 \\ 0 & 2 & 2 \\ -4 & 0 & -1 \end{pmatrix}$
- e) Symmetric square matrices: $\begin{pmatrix} 0 & 1 \\ 1 & -3 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & -4 \\ 0 & 2 & 2 \\ -4 & 2 & -1 \end{pmatrix}$
- f) Diagonal square matrices: $\begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
- g) Identity matrices: $\mathbf{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\mathbf{I}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- h) Upper triangular matrices: $\begin{pmatrix} 1 & 4 \\ 0 & -2 \end{pmatrix}$ $\begin{pmatrix} 1 & 5 & -3 \\ 0 & -2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$
- i) Lower triangular matrices: $\begin{pmatrix} 1 & 0 \\ 3 & -2 \end{pmatrix}$ $\begin{pmatrix} -5 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 0 & -3 \end{pmatrix}$

3.3. Operations with matrices. Inverse matrix**Sum of matrices**

In order to sum (or subtract) two matrices $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$, both of them must have the same dimension. In this case, we simply add the terms in the same position, i.e.:

$$\mathbf{A} + \mathbf{B} = \mathbf{C} = (c_{ij}) \quad \text{where } c_{ij} = a_{ij} + b_{ij}, \forall i, j$$

Example 20

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} a+1 & b+2 & c+3 \\ d+4 & e+5 & f+6 \end{pmatrix}$$

The sum of matrices verifies the following properties:

- 1) **Associative:** $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$.
- 2) **Commutative:** $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
- 3) **Identity element:** $\mathbf{A} + \mathbf{0} = \mathbf{0} + \mathbf{A} = \mathbf{A}$.
- 4) **Inverse:** $\mathbf{A} + (-\mathbf{A}) = (-\mathbf{A}) + \mathbf{A} = \mathbf{0}$, for $\mathbf{A} = (a_{ij})$ and $-\mathbf{A} = (-a_{ij})$.

Product of a scalar by a matrix

In order to multiply a number $k \in \mathbb{R}$ by a matrix $\mathbf{A} = (a_{ij})$, we simply multiply the number by every element of the matrix, i.e.:

$$k \cdot \mathbf{A} = \mathbf{C} = (c_{ij}) \quad \text{where } c_{ij} = k \cdot a_{ij}, \forall i, j$$

Example 21

$$k \cdot \begin{pmatrix} 1 & 0 \\ 3 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} k \cdot 1 & k \cdot 0 \\ k \cdot 3 & k \cdot 1 \\ k \cdot (-1) & k \cdot 2 \end{pmatrix}$$

The product of a scalar by a matrix verifies the following properties:

- 1) $k \cdot (\mathbf{A} + \mathbf{B}) = k \cdot \mathbf{A} + k \cdot \mathbf{B}$.
- 2) $(k + h) \cdot \mathbf{A} = k \cdot \mathbf{A} + h \cdot \mathbf{A}$.
- 3) $k \cdot (h \cdot \mathbf{A}) = (k \cdot h) \cdot \mathbf{A}$.
- 4) Identity element: $1 \cdot \mathbf{A} = \mathbf{A} \cdot 1 = \mathbf{A}$.

For any $k, h \in \mathbb{R}$

Matrix product

In order to multiply a matrix $\mathbf{A} = (a_{ij})$ with another $\mathbf{B} = (b_{ij})$, we need the number of columns of $\mathbf{A}$ to be equal to the number of rows of $\mathbf{B}$. In this case, the result is a new matrix $\mathbf{C} = (c_{ij})$ with as many rows as $\mathbf{A}$ and as many columns as $\mathbf{B}$, where its elements are computed as:

$$c_{ij} = a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + \dots + a_{in} \cdot b_{nj}$$

Example 22

If $\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 & -1 \\ 3 & -2 & 2 \\ 3 & -3 & 1 \end{pmatrix}$, we can multiply $\mathbf{A}$ by $\mathbf{B}$, since the number of columns of $\mathbf{A}$ coincides with the number of rows of $\mathbf{B}$. The result is a matrix of dimension 2×3 :

$$\mathbf{A} \cdot \mathbf{B} = \begin{pmatrix} a \cdot 0 + b \cdot 3 + c \cdot 3 & a \cdot 1 + b \cdot (-2) + c \cdot (-3) & a \cdot (-1) + b \cdot 2 + c \cdot 1 \\ d \cdot 0 + e \cdot 3 + f \cdot 3 & d \cdot 1 + e \cdot (-2) + f \cdot (-3) & d \cdot (-1) + e \cdot 2 + f \cdot 1 \end{pmatrix}$$

On the other hand, we simply cannot compute the product of $\mathbf{B}$ by $\mathbf{A}$, since the number of columns of $\mathbf{B}$ is not the same as the number of rows of $\mathbf{A}$.

The matrix product verifies the following properties:

- 1) **Associative:** $(\mathbf{A} \cdot \mathbf{B}) \cdot \mathbf{C} = \mathbf{A} \cdot (\mathbf{B} \cdot \mathbf{C})$.
- 2) It is **not commutative**: $\mathbf{A} \cdot \mathbf{B} \neq \mathbf{B} \cdot \mathbf{A}$.
- 3) If $\mathbf{A}$ is a square matrix of order n , it holds $\mathbf{A} \cdot \mathbf{I}_n = \mathbf{I}_n \cdot \mathbf{A} = \mathbf{A}$ for $\mathbf{I}_n$ the identity matrix of order n .
- 4) If $\mathbf{A}$ is a square matrix of order n , a matrix $\mathbf{B}$ such that $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} = \mathbf{I}_n$ **not always** exists. Whenever such a $\mathbf{B}$ exists, it will be called **inverse matrix** of $\mathbf{A}$ and will be written as $\mathbf{A}^{-1}$. A matrix without inverse is called **singular matrix**.
- 5) Distributive with respect to the sum: $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$, and $(\mathbf{A} + \mathbf{B}) \cdot \mathbf{C} = \mathbf{A} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{C}$.

4. Determinants

In this chapter we will present the concept of determinant of a matrix, diverse techniques for computing it and important applications to the study of matrices and vector spaces.

4.1. Determinant of a square matrix of order 2 or 3

Every square matrix of order $n \geq 1$ can be associated to a particular number, which we will call **determinant of A** and is usually written as $\det(\mathbf{A})$ or simply $|\mathbf{A}|$.

- If $n = 1$, $\mathbf{A} = (a_{11})$ and $|\mathbf{A}| = |a_{11}| = a_{11}$.
- If $n = 2$, $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, and $|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11} \cdot a_{22} - a_{12} \cdot a_{21}$.
- If $n = 3$, $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, and

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \cdot a_{22} \cdot a_{33} + a_{21} \cdot a_{32} \cdot a_{13} + a_{31} \cdot a_{23} \cdot a_{12} - a_{13} \cdot a_{22} \cdot a_{31} - a_{23} \cdot a_{32} \cdot a_{11} - a_{33} \cdot a_{21} \cdot a_{12}$$

Note

A matrix, which is a table of numbers, must not be confused with its determinant, which is a single number.

Observation

The numbers of a matrix are enclosed by parenthesis, but those of a determinant are enclosed between two vertical bars.

The well-known Sarrus rule is a good way to memorize the computations for a determinant of order 3, and it is represented by:

Figure 3. Sarrus rule for the calculus of the determinant of a matrix of order 3

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

For the cases $n > 3$ the computation of the determinant isn't as immediate as those, since they need some additional concepts that will be explained in the following page.

Example 23

a) Order 2 matrices:

$$\begin{vmatrix} 1 & -2 \\ 0 & 3 \end{vmatrix} = 1 \cdot 3 - 0 \cdot (-2) = 3$$

$$\begin{vmatrix} 2 & 5 \\ 3 & -4 \end{vmatrix} = 2 \cdot (-4) - 5 \cdot 3 = -23$$

b) Order 3 matrices:

$$\begin{vmatrix} 2 & 3 & 1 \\ -2 & -1 & 2 \\ 0 & -1 & 1 \end{vmatrix} = 2 \cdot (-1) \cdot 1 + 3 \cdot 2 \cdot 0 + 1 \cdot (-2) \cdot (-1) - 0 \cdot (-1) \cdot 1 - 1 \cdot (-2) \cdot 3 - 2 \cdot (-1) \cdot 2 = 10$$

$$\begin{vmatrix} 1 & 4 & -3 \\ 3 & -2 & 1 \\ 4 & -1 & 5 \end{vmatrix} = 1 \cdot (-2) \cdot 5 + 4 \cdot 4 \cdot 1 + 3 \cdot (-1) \cdot (-3) - 4 \cdot (-2) \cdot (-3) - 5 \cdot 3 \cdot 4 - 1 \cdot (-1) \cdot 1 = 10 + 9 + 16 - (24 - 1 + 60) = 15 - 83 = -68$$

4.2. Determinant of a square matrix of order 4 or higher

When we want to compute the determinant of a square matrix of order $n > 3$, we will need to use the concept of adjugate of an element of the matrix:

Let

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

be a square matrix. We call **adjugate** or **adjoint** of an element a_{ij} , denoted by A_{ij} to the following number:

- a) The resulting determinant of deleting the row i and the column j of the matrix $\mathbf{A}$.
- b) The following sign in front of such determinant:

$$\begin{cases} + & \text{if } i + j \text{ is even} \\ - & \text{if } i + j \text{ is odd} \end{cases}$$

In other words, this sign will be $(-1)^{i+j}$.

Observe that the sign associated to every adjugate depends exclusively on the position of the element a_{ij} in the matrix $\mathbf{A}$ in such a way that:

$$\begin{vmatrix} + & - & + & - & \dots \\ - & + & - & + & \dots \\ + & - & + & - & \dots \\ - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{vmatrix}$$

Example 24

Given a square matrix of order 3, $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, the adjoint associated to the element a_{21} can be obtained such as:

- 1) We consider the determinant after deleting row 2 and column 1:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- 2) Compute the sign depending on the position. In this case $2 + 1 = 3$, which is odd. Therefore, the sign will be negative.

Therefore:

$$A_{21} = - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}.$$

Example 25

Given a square matrix of order $n = 4$, $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$.

We have:

$$A_{11} = (-1)^{1+1} \cdot \begin{vmatrix} a_{22} & a_{23} & a_{24} \\ a_{32} & a_{33} & a_{34} \\ a_{42} & a_{43} & a_{44} \end{vmatrix}, \quad A_{12} = (-1)^{1+2} \cdot \begin{vmatrix} a_{21} & a_{23} & a_{24} \\ a_{31} & a_{33} & a_{34} \\ a_{41} & a_{43} & a_{44} \end{vmatrix} \dots$$

Proposition. We can compute the determinant of a square matrix $\mathbf{A}$ of order n using the adjugates of a single row or column.

Indeed, using the adjugates of the elements in row i :

$$|\mathbf{A}| = a_{i1} \cdot A_{i1} + a_{i2} \cdot A_{i2} + \dots + a_{in} \cdot A_{in}.$$

Also, using the adjugates of the elements in column j :

$$|\mathbf{A}| = a_{1j} \cdot A_{1j} + a_{2j} \cdot A_{2j} + \dots + a_{nj} \cdot A_{nj}.$$

Example 26

Let's compute, using adjugates, the determinant of the following matrix of order 3:

$$\begin{pmatrix} 2 & 3 & 1 \\ -2 & -1 & 2 \\ 0 & -1 & 1 \end{pmatrix}$$

Developing the adjoints of the first row:

$$\begin{vmatrix} 2 & 3 & 1 \\ -2 & -1 & 2 \\ 0 & -1 & 1 \end{vmatrix} = 2 \cdot \begin{vmatrix} -1 & 2 \\ -1 & 1 \end{vmatrix} - 3 \cdot \begin{vmatrix} -2 & 2 \\ 0 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} -2 & -1 \\ 0 & -1 \end{vmatrix} = 2 \cdot (1) - 3 \cdot (-2) + 1 \cdot (2) = 10.$$

Alternatively, developing by the adjoints of the first column

$$\begin{vmatrix} 2 & 3 & 1 \\ -2 & -1 & 2 \\ 0 & -1 & 1 \end{vmatrix} = 2 \cdot \begin{vmatrix} -1 & 2 \\ -1 & 1 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 3 & 1 \\ -1 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 3 & 1 \\ -1 & 2 \end{vmatrix} = 2 \cdot (1) + 2 \cdot (4) + 0 \cdot (7) = 10.$$

Example 27

Let's compute the determinant of the matrix of order 4:

$$\begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 \\ 2 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \end{pmatrix}$$

We will develop using the adjoints of the second row (it is usually convenient to choose the row or columns with more zeroes, since the computations will be easier):

$$\begin{vmatrix} 1 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 \\ 2 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \end{vmatrix} = 0 \cdot A_{21} + 0 \cdot A_{22} - 1 \cdot A_{23} + 0 \cdot A_{24} = -A_{23}$$

where

$$A_{23} = (-1)^{2+3} \cdot \begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 2 & 1 \end{vmatrix} = -3.$$

Therefore:

$$\begin{vmatrix} 1 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 \\ 2 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \end{vmatrix} = -(-3) = 3.$$

4.3. Properties of determinants

The following properties can be useful when computing determinants of any order:

- 1) The determinant of a matrix is the same as the determinant of the transposed matrix.

Example 28

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 3 & 0 & 1 \end{vmatrix} = -2$$

- 2) Exchanging the positions of two rows (or columns) of a matrix changes the sign of it (but not the absolute value).

Example 29

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = - \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

Indeed: $\begin{vmatrix} 1 & 2 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = -2$ and $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} = 2$.

- 3) If two rows (or columns) of a matrix are equal or proportionals one to the other, its determinant is 0.

Example 30

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 0 & 0 \end{vmatrix} = 0 \text{ since the two first rows (R1 and R2) are the same.}$$

Note

R1 means *row 1*.
C1 means *column 1*.

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \end{vmatrix} = 0 \text{ since the two first columns (C1 and C2) are proportional.}$$

4) A determinant that has a null row or column is of value 0.

Example 31

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 4 & 3 \end{vmatrix} = 0 \text{ since the second row R2 is null.}$$

5) Multiplying a determinant by a number is equivalent to multiply any row or column of the determinant by such a number.

Example 32

$$2 \cdot \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 2 \cdot 3 \\ 0 & 0 & 2 \cdot 1 \\ 1 & 0 & 2 \cdot 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 2 \cdot 0 & 2 \cdot 0 & 2 \cdot 1 \\ 1 & 0 & 0 \end{vmatrix}$$

6) If all the elements of the same row or column of a matrix are written as a sum, its determinant can be decomposed as the sum of two determinants of matrices exchanging that row or column by the two components of the sum.

Example 33

$$\begin{vmatrix} 2 & 4+5 & 3 \\ 0 & 3+3 & 2 \\ 0 & 1+2 & 5 \end{vmatrix} = \begin{vmatrix} 2 & 4 & 3 \\ 0 & 3 & 2 \\ 0 & 1 & 5 \end{vmatrix} + \begin{vmatrix} 2 & 5 & 3 \\ 0 & 3 & 2 \\ 0 & 2 & 5 \end{vmatrix}$$

7) Whenever any row or column of a matrix is a linear combination of the others, its determinant is 0.

Example 34

$$\begin{vmatrix} 1 & 3+2 \cdot 1 & 3 \\ 2 & 0+2 \cdot 2 & 0 \\ 3 & 1+2 \cdot 3 & 1 \end{vmatrix} = 0 \text{ since C2 is a linear combination of C1 and C3.}$$

- 8) If we add a linear combination of the other rows (or columns) of a matrix to a particular row (or column), the determinant doesn't change.

Example 35

$$\begin{vmatrix} 1 & 5 & 3 \\ 2 & 3 & 0 \\ 3 & 4 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 5+2 \cdot 1 & 3 \\ 2 & 3+2 \cdot 2 & 0 \\ 3 & 4+2 \cdot 3 & 1 \end{vmatrix}$$

- 9) The determinant of the product of two matrices is exactly the product of their determinants.

Example 36

Let $A = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Then $A \cdot B = \begin{pmatrix} 0 & 2 \\ 1 & 4 \end{pmatrix}$, $|A| = 2$, $|B| = -1$, $|A \cdot B| = -2$, so $|A \cdot B| = |A| \cdot |B|$.

4.4. Computation of the inverse matrix

When we studied the properties of the product of two matrices, we said:

- a) If $\mathbf{A}$ is a square matrix of order n , **not** always exists a matrix $\mathbf{B}$ such that $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} = \mathbf{I}_n$ (where $\mathbf{I}_n$ is the identity matrix of order n).
- b) Whenever such a $\mathbf{B}$ exists, it will be called **inverse matrix** of $\mathbf{A}$ and will be written as $\mathbf{A}^{-1}$.
- c) A matrix without inverse is called **singular matrix**.

Proposition. A square matrix $\mathbf{A}$ is non-singular (i.e., $\mathbf{A}$ has inverse) if and only if $\det(\mathbf{A}) \neq 0$. In this case, it holds:

$$\mathbf{A}^{-1} = \frac{(\text{adj}(\mathbf{A}))^t}{\det(\mathbf{A})},$$

where $\text{adj}(\mathbf{A})$ is the matrix obtained after exchanging every element of A by its adjugate.

Example 37

The matrix $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ has inverse, since $|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = -4 \neq 0$.

Let's compute the adjoints of the elements in $\mathbf{A}$ (taking into account the corresponding sign):

$$\begin{aligned} A_{11} &= \begin{vmatrix} -1 & 0 \\ 1 & 1 \end{vmatrix} = -1 & A_{12} &= -\begin{vmatrix} -2 & 0 \\ 0 & 1 \end{vmatrix} = 2 & A_{13} &= \begin{vmatrix} -2 & -1 \\ 0 & 1 \end{vmatrix} = -2 \\ A_{21} &= -\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = 1 & A_{22} &= \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} = 2 & A_{23} &= -\begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} = -2 \\ A_{31} &= \begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} = 1 & A_{32} &= -\begin{vmatrix} 2 & 1 \\ -2 & 0 \end{vmatrix} = -2 & A_{33} &= \begin{vmatrix} 2 & 0 \\ -2 & 1 \end{vmatrix} = -2 \end{aligned}$$

Therefore $\text{adj}(\mathbf{A}) = \begin{pmatrix} -1 & 2 & -2 \\ 1 & 2 & -2 \\ 1 & -2 & -2 \end{pmatrix}$.

And $\mathbf{A}^{-1} = \frac{(\text{adj}(\mathbf{A}))^t}{|\mathbf{A}|} = \frac{1}{-4} \begin{pmatrix} -1 & 1 & 1 \\ 2 & 2 & -2 \\ -2 & -2 & -2 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{-1}{4} & \frac{-1}{4} \\ \frac{-1}{2} & \frac{-1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$

It can easily be proved that $\mathbf{A} \cdot \mathbf{A}^{-1} = \mathbf{A}^{-1} \cdot \mathbf{A} = \mathbf{I}_3$:

$$\mathbf{A} \cdot \mathbf{A}^{-1} = \begin{pmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{4} & \frac{-1}{4} & \frac{-1}{4} \\ \frac{-1}{2} & \frac{-1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{A}^{-1} \cdot \mathbf{A} = \begin{pmatrix} \frac{1}{4} & \frac{-1}{4} & \frac{-1}{4} \\ \frac{-1}{2} & \frac{-1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

4.5. Rank of a matrix. Computation using determinants

The **rank of a matrix** is the number of rows or columns linearly independent. If the matrix is called $\mathbf{A}$, its rank is denoted by $\text{rk}(\mathbf{A})$.

Note

It can be proved that, in any matrix, the number of linearly independent rows coincides with the number of linearly independent columns.

Example 38

Given the matrix $\mathbf{A} = \begin{pmatrix} 1 & -2 & 4 & 2 \\ 0 & 3 & 1 & 1 \\ 0 & 0 & -2 & 3 \end{pmatrix}$ the number of linearly independent rows is 3, therefore its rank is also 3.

In the previous example it was easy to compute the rank of the matrix, since it is upper triangular. But usually the computation of the rank of any given matrix isn't as easy.

The concept of matrix has different applications in linear algebra and geometry, so it is extremely important to develop a method to compute efficiently the rank of a given matrix.

Given a matrix $\mathbf{A}$ with m rows and n columns, we call **minor of order h** (for $1 \leq h \leq \min(m, n)$) to any determinant obtained after selecting only h rows and h columns from the matrix $\mathbf{A}$.

Example 39. Minor of a matrix

Given the matrix $\mathbf{A} = \begin{pmatrix} 2 & -3 & 4 & 1 \\ -4 & 6 & -8 & -2 \\ 2 & 0 & 1 & 0 \end{pmatrix}$, a minor of order 2 is obtained after selecting two rows and two columns. We can select the two last rows and the two last columns, and get the following minor

$$\begin{vmatrix} -8 & -2 \\ 1 & 0 \end{vmatrix} = 2$$

Another different possible minor can be obtained selecting the last two rows and the first and third columns:

$$\begin{vmatrix} -4 & -8 \\ 2 & 1 \end{vmatrix} = 12$$

Given a certain minor of order h , we call **bordered minor** to the determinant obtained after adding to the minor, in an ordered way, the elements of a new row and column.

Note

This means that we can only add elements that, in the original matrix, are in the same row or column that the elements of the minor.

Example 40. Bordered minor

Following the previous example, from the last minor we can generate two different bordered minors:

- The minor of order 3 obtained adding the first row and the last column:

$$\begin{vmatrix} 2 & 4 & 1 \\ -4 & -8 & -2 \\ 2 & 1 & 0 \end{vmatrix}$$

- The minor of order 3 obtained adding the first row and the second column:

$$\begin{vmatrix} 2 & -3 & 4 \\ -4 & 6 & -8 \\ 2 & 0 & 1 \end{vmatrix}$$

After introducing the concept of minor and bordered minor, we should highlight the following key result in order to compute the rank of a matrix:

Rank's theorem. The rank of a non-null matrix is given by its larger non-null minor.

In other words: if all the elements of the matrix are null, the rank of the matrix is zero. If the matrix has any element different than zero, its rank will be equal or greater than one. In order to find the rank, we will need to find the larger non-null minor that can be formed with the matrix elements. The order of such a minor will be the rank of the matrix.

Example 41. Computation of a matrix's rank

Using the matrix from the previous examples, it is obvious that its rank is larger than 1 (because the matrix isn't null) and smaller or equal than 3 (because the larger minor we can take from it will have order 3).

It is easy to prove that the rank of the matrix will be greater or equal than 2, since the following minor of order 2 is different from zero:

$$\begin{vmatrix} -8 & -2 \\ 1 & 0 \end{vmatrix} = 2 \neq 0$$

We will use the previous minor as a “basis of order 2” in order to get any minor of order 3 different from 0. If we cannot find any using this minor as a “basis”, we won't be able to using any other either.

Nevertheless, the only two minors of order 3 we can get from the previous minor of order two are the following:

$$\begin{vmatrix} 2 & 4 & 1 \\ -4 & -8 & -2 \\ 2 & 1 & 0 \end{vmatrix} = 0, \quad \begin{vmatrix} -3 & 4 & 1 \\ 6 & -8 & -2 \\ 0 & 1 & 0 \end{vmatrix} = 0$$

We cannot find any non-null minor of order 3. Therefore, the rank of this matrix is 2, i.e.: $\mathbf{rk}(\mathbf{A}) = 2$.

Example 42. Computation of a matrix's rank

We want to find the rank of the following matrix:

$$\mathbf{A} = \begin{pmatrix} \boxed{\begin{matrix} 2 & 3 \\ 1 & 0 \end{matrix}} & 0 & 2 \\ 4 & 6 & 0 & 4 \\ 2 & 3 & 0 & 1 \end{pmatrix}$$

Obviously, the rank will be between the numbers 1 (because the matrix isn't null) and 4 (because, at most, we will be able to take a minor of order 4).

It is easy to find a minor of order 2 different from zero: $\begin{vmatrix} 2 & 3 \\ 1 & 0 \end{vmatrix} = -3 \neq 0$. Therefore, it must be $\mathbf{rk}(\mathbf{A}) \geq 2$.

The following step will be taking bordered minors after the previous one (our “basis of order 2”) in order to look for a possible minor of order 3 different from 0:

$$\begin{vmatrix} 2 & 3 & 0 \\ 1 & 0 & 1 \\ 4 & 6 & 0 \end{vmatrix} = 0, \text{ since } F3 \text{ is a multiple of } F1.$$

$$\begin{vmatrix} 2 & 3 & 0 \\ 1 & 0 & 1 \\ 2 & 3 & 0 \end{vmatrix} = 0, \text{ since } F3 \text{ is exactly } F1.$$

$$\begin{vmatrix} 2 & 3 & 2 \\ 1 & 0 & 1 \\ 2 & 3 & 1 \end{vmatrix} = 3 \neq 0.$$

We have a minor of order 3 different from 0, therefore $\mathbf{rk}(\mathbf{A}) \geq 3$. We will use this minor as a “basis of order 3”.

However, we have $\mathbf{rk}(\mathbf{A}) \neq 0$ since the only minor of order 4 we have in this matrix is 0:

$$\begin{vmatrix} 2 & 3 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 4 & 6 & 0 & 4 \\ 2 & 3 & 0 & 1 \end{vmatrix} = 0 \text{ because } F3 \text{ is a multiple of } F1.$$

Therefore, $\mathbf{rk}(\mathbf{A}) = 3$.

4.6. Applications to vector spaces

1) Linear dependence or independence

Let $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a set of vectors in the vector space V , with $\mathbf{dim}(V) = n$. Sorting them as rows or columns we get a matrix $\mathbf{A}$.

We have $\mathbf{v}_1, \dots, \mathbf{v}_k$ linearly independent if and only if $\mathbf{rk}(\mathbf{A}) = k$. Therefore, $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly dependent if and only if $\mathbf{rk}(\mathbf{A}) < k$.

Example 43

The vectors $(2, 1, 2)$, $(3, 2, 1)$, $(-1, 1, -7)$ of $\mathbb{R}^3$ are linearly dependent, since the matrix that has them as columns $\mathbf{A} = \begin{pmatrix} 2 & 3 & -1 \\ 1 & 2 & 1 \\ 2 & 1 & -7 \end{pmatrix}$ has $\mathbf{rk}(\mathbf{A}) = 2$ (it is enough to check that $|\mathbf{A}| = 0$ and that there's a minor of order 2 different from 0).

Proposition. A set of n vectors is linearly independent in $\mathbb{R}^n$ if and only if the determinant of the matrix build with them is different than 0.

Example 44

Let's use the previous proposition to prove that the rank of the set:

$$\{(1, 3, 0), (-1, 2, -4), (1, 1, 2)\}$$

is 3, i.e., the three vectors are linearly independent:

$$\begin{vmatrix} 1 & 3 & 0 \\ -1 & 2 & -4 \\ 1 & 1 & 2 \end{vmatrix} = 4 - 12 - (-4) - (-6) = 2 \neq 0$$

Example 45

Let's use the previous proposition to prove that the rank of the set

$$\{(1, 3, 0), (-1, 2, -4), (0, 5, -4)\}$$

is less than 3, i.e.: the three vectors are linearly dependent:

$$\begin{vmatrix} 1 & 3 & 0 \\ -1 & 2 & -4 \\ 0 & 5 & -4 \end{vmatrix} = -8 - (-20) - 12 = 0$$

2) Dimension of a generated subspace

Let $W = \langle \mathbf{v}_1, \dots, \mathbf{v}_k \rangle$ a generated subspace by a set of vectors. Let $\mathbf{A}$ be the matrix that has these vectors as rows (or columns), it holds:

$$\dim(W) = \text{rk}(\mathbf{A}).$$

Example 46

The vectors $(1, 2, 3, 4), (5, 6, 7, 8), (4, 4, 4, 4)$ of $\mathbb{R}^4$ define the matrix: $\begin{pmatrix} 1 & 5 & 4 \\ 2 & 6 & 4 \\ 3 & 7 & 4 \\ 4 & 8 & 4 \end{pmatrix}$.

It can be proven that $\text{rk}(\mathbf{A}) = 2$ (use software to do it), therefore these three vectors generate a vector subspace of $\mathbb{R}^4$ with dimension 2.

4.7. Change-of-basis matrix of a vector space

Remember that a basis B of a vector space V , of $\dim(V) = n$, give a system of coordinates for V and every vector $\mathbf{v} \in V$ can be identified uniquely with its components in this basis (real numbers $c_1, \dots, c_n$).

In order to get a better understanding, we will identify $(c_1, \dots, c_n)$ as a vector of $\mathbb{R}^n$ (or a matrix $1 \times n$).

Let $B = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ and $A = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ two basis of a vector space V .

Let $\mathbf{C}$ be the square matrix of order n whose columns are the components of the vectors of B in the basis A , $\mathbf{C} = (c_{ij})_{n \times n}$ is called the **change-of-basis matrix** from B to A .

If a vector $\mathbf{v} \in V$ has components $b_1, \dots, b_n$ in the basis B and $a_1, \dots, a_n$ in the basis A , it holds:

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} c_{11} & \dots & c_{1n} \\ \vdots & & \vdots \\ c_{n1} & \dots & c_{nn} \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

Example 47

Let $B = \{(1, 1, 1), (1, 0, 3), (3, 4, 5)\}$ and $A = \{(2, 3, -1), (0, 0, 1), (2, 1, 0)\}$ two basis of $\mathbb{R}^3$. In order to compute the change-of-basis matrix from B to A we follow the following steps:

- 1) Compute the components of $(1, 1, 1)$ in A solving the matrix equation:

$$\underbrace{\begin{pmatrix} 2 & 0 & 2 \\ 3 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}}_{\mathbf{M}} \cdot \begin{pmatrix} c_{11} \\ c_{21} \\ c_{31} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Since the determinant of this square matrix is different than 0, the matrix can be inverted. Therefore:

$$\begin{pmatrix} c_{11} \\ c_{21} \\ c_{31} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 3 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

- 2) In the same way, we compute the components of the other two vectors:

$$\begin{pmatrix} c_{12} \\ c_{22} \\ c_{32} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 3 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} c_{13} \\ c_{23} \\ c_{33} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 2 \\ 3 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}.$$

It could have been done in a single step with the following matrix operation:

$$\begin{pmatrix} 2 & 0 & 2 \\ 3 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 & 1 & 3 \\ 1 & 0 & 4 \\ 1 & 3 & 5 \end{pmatrix} = \underbrace{\begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}}_{\mathbf{C}}$$

where the result is the change-of-basis matrix $\mathbf{C}$ from B to A .

Once we have this change-of-basis matrix we can simply use it as previously explained. As an example, a vector that has $(1, 1, 0)$ as components in the basis B will have the following components in the basis A :

$$\mathbf{C} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}.$$

5. Equations of lines and planes

5.1. Equations of a line in the plane

It is intuitive to see that any two different points in the plane $\mathbb{R}^2$ define uniquely a line r passing through both of them. A line in the plane can also be defined by a given point and a vector giving the direction of the line (director vector). As we will see in the following sections, the equations of a line can be written in different equivalent expressions.

Vectorial equation

Given two different points, $P : (p_1, p_2)$ and $Q : (q_1, q_2)$, of the line r , we can take the vector $\mathbf{v} = \overrightarrow{PQ} = (v_1, v_2) = (q_1 - p_1, q_2 - p_2)$. Any other point $X : (x, y)$ will verify the equation:

$$X = P + k \cdot \mathbf{v}, \text{ for } k \in \mathbb{R} \quad (1)$$

Parametric equation

The equation (1) can be written as follows:

$$(x, y) = (p_1, p_2) + k \cdot (v_1, v_2),$$

This is:

$$(x, y) = (p_1 + k \cdot v_1, p_2 + k \cdot v_2)$$

Which is equivalent to the following system of equations:

$$\begin{cases} x &= p_1 + k \cdot v_1 \\ y &= p_2 + k \cdot v_2 \end{cases} \quad k \in \mathbb{R} \quad (2)$$

Continuous equation

After isolating the parameter k in (2), we get the following equations of r (whenever $v_1, v_2 \neq 0$ and $v_1 \neq v_2$):

$$\begin{cases} x &= p_1 + k \cdot v_1 \\ y &= p_2 + k \cdot v_2 \end{cases} \Leftrightarrow \begin{cases} k &= \frac{x-p_1}{v_1} \\ k &= \frac{y-p_2}{v_2} \end{cases}$$

Therefore:

$$\frac{x-p_1}{v_1} = \frac{y-p_2}{v_2} \quad (3)$$

Slope-intercept equation and explicit equation

Isolating y from the previous expression (3) we get:

1) The equation slope-intercept $y - p_2 = \frac{v_2}{v_1}(x - p_1)$, which is:

$$y - p_2 = m \cdot (x - p_1), \quad (4)$$

where $m = \frac{v_2}{v_1}$ is the **slope** of r .

2) The explicit equation $y = m \cdot (x - p_1) + p_2 = m \cdot x - m \cdot p_1 + p_2$, which is:

$$y = m \cdot x + n, \quad (5)$$

where $n = -m \cdot p_1 + p_2$ is the **origin ordinate**.

General equation

We can also develop the expression (3) as follows:

$$\frac{x-p_1}{v_1} = \frac{y-p_2}{v_2} \Leftrightarrow v_2 \cdot (x-p_1) = v_1 \cdot (y-p_2) \Leftrightarrow v_2 \cdot x - v_1 \cdot y - v_2 \cdot p_1 + v_1 \cdot p_2 = 0$$

Or, written with easier notation:

$$Ax + By + C = 0, \quad (6)$$

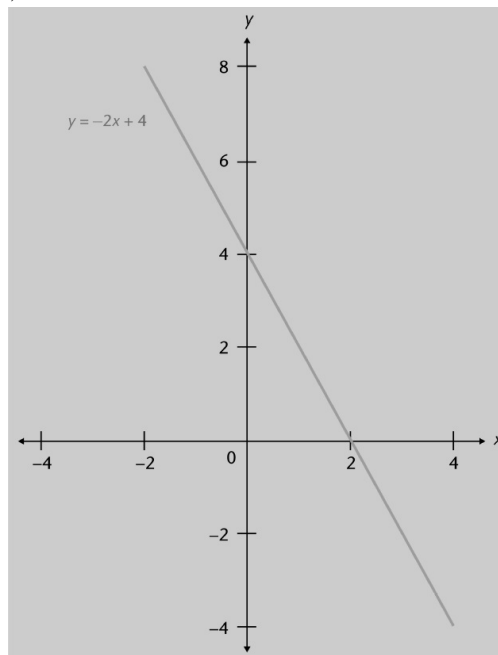
for $A = v_2$, $B = -v_1$ and $C = -v_2 \cdot p_1 + v_1 \cdot p_2$.

Example 48

Let r be the line through $P : (1, 2)$ with direction $\mathbf{v} = (1, -2)$.

- Vectorial equation: $(x, y) = (1, 2) + k \cdot (1, -2)$ for $k \in \mathbb{R}$.
- Parametric equation: $\begin{cases} x = 1 + k \cdot 1 \\ y = 2 + k \cdot (-2) \end{cases}$ for $k \in \mathbb{R}$.
- Continuous equation: $\frac{x-1}{1} = \frac{y-2}{-2}$.
- Explicit equation: $y = -2x + 4$.
- General equation: $2x + y - 4 = 0$.

Figure 4. Graphical representation of the line that passes through $P : (1, 2)$ with director vector $\mathbf{v} = (1, -2)$



5.2. Equations of a line in the space

In such a similar way of what happens in $\mathbb{R}^2$, we can also consider different alternative expressions for representing a line with equations in $\mathbb{R}^3$ (the development of them is analogous to $\mathbb{R}^2$):

Let r be a line, $P : (p_1, p_2, p_3)$ and $Q : (q_1, q_2, q_3)$ two points belonging to r and the director vector $\mathbf{v} = \overrightarrow{PQ} = (v_1, v_2, v_3)$. Any other point $X : (x, y, z)$ of the line r satisfies the equation $X = P + k \cdot \mathbf{v}$ for $k \in \mathbb{R}$.

There are also the following alternative equations (whenever $v_1 \cdot v_2 \cdot v_3 \neq 0$):

- **Vectorial equation:** $(x, y, z) = (p_1, p_2, p_3) + k \cdot (v_1, v_2, v_3)$.

- **Parametric equations:**
$$\begin{cases} x = p_1 + k \cdot v_1 \\ y = p_2 + k \cdot v_2 \\ z = p_3 + k \cdot v_3 \end{cases}$$

- **Continuous equations:** $\frac{x - p_1}{v_1} = \frac{y - p_2}{v_2} = \frac{z - p_3}{v_3}$

Example 49

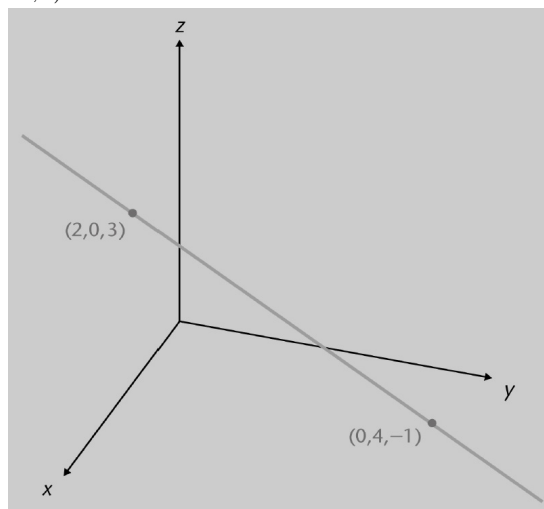
Let r be the line through $P : (1, 2, 1)$ and whose direction is given by $\mathbf{v} = (1, -2, 2)$.

- Vectorial equation: $(x, y, z) = (1, 2, 1) + k \cdot (1, -2, 2)$ for $k \in \mathbb{R}$.

- Parametric equations:
$$\begin{cases} x = 1 + k \\ y = 2 - 2k \\ z = 1 + 2k \end{cases} \text{ for } k \in \mathbb{R}.$$

- Continuous equations: $\frac{x - 1}{1} = \frac{y - 2}{-2} = \frac{z - 1}{2}.$

Figure 5. Graphical representation of the line that passes through $P : (1, 2, 1)$ with director vector $\mathbf{v} = (1, -2, 2)$



5.3. Equations of a plane in the space

The equation of a plane π in the space $\mathbb{R}^3$ is defined by a point of the plane $P : (p_1, p_2, p_3)$ and two vectors $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$ different than zero and linearly independent (i.e. not parallel). It is possible to consider different equivalent expressions for the equation of a plane in the space:

- **Vectorial equation:**

$$(x, y, z) = (p_1, p_2, p_3) + k \cdot (u_1, u_2, u_3) + h \cdot (v_1, v_2, v_3) \text{ for } k, h \in \mathbb{R}.$$

- **Parametric equations:**

$$\begin{cases} x &= p_1 + k \cdot u_1 + h \cdot v_1 \\ y &= p_2 + k \cdot u_2 + h \cdot v_2 \\ z &= p_3 + k \cdot u_3 + h \cdot v_3 \end{cases} \text{ for } k, h \in \mathbb{R}.$$

- **General equation:** From the equation

$$\begin{vmatrix} x - p_1 & y - p_2 & z - p_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = 0$$

we can obtain an expression $Ax + By + Cz + D = 0$.

Example 50

Let π be the plane through the point $P : (1, 2, 3)$ and with director vectors $\mathbf{u} = (1, 0, -1)$, $\mathbf{v} = (0, 1, 1)$.

We have:

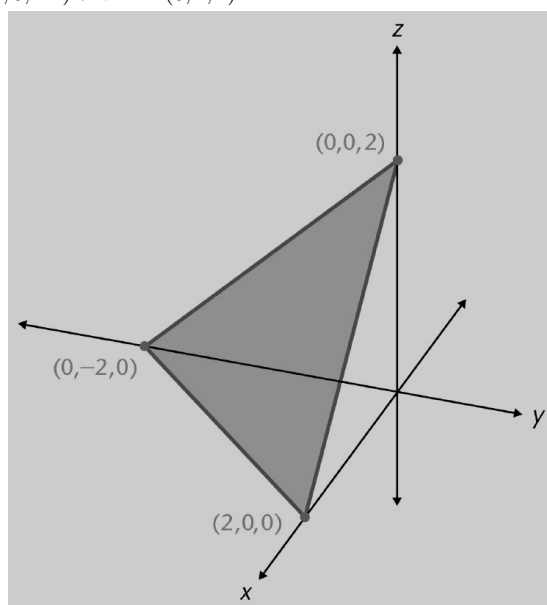
- Vectorial equation: $(x, y, z) = (1, 2, 3) + k \cdot (1, 0, -1) + h \cdot (0, 1, 1)$ for $k, h \in \mathbb{R}$.

- Parametric equations: $\begin{cases} x &= 1 + k \\ y &= 2 + h \\ z &= 3 - k + h \end{cases} \text{ for } k, h \in \mathbb{R}.$

- General equation:

$$\begin{vmatrix} x - 1 & y - 2 & z - 3 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 0$$

Figure 6. Graphical representation of the plane passing through $P: (1, 2, 3)$ with director vectors $\mathbf{u} = (1, 0, -1)$ and $\mathbf{v} = (0, 1, 1)$



6. Scalar product and orthogonality

6.1. Scalar product, norm of a vector and angle between two vectors

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$, $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be two vectors of the vector space $\mathbb{R}^n$. We define the **scalar product** of them, $\mathbf{u} \cdot \mathbf{v}$, as follows:

$$\mathbf{u} \cdot \mathbf{v} = u_1 \cdot v_1 + u_2 \cdot v_2 + \dots + u_n \cdot v_n$$

Observe that:

- a) The scalar product of two vectors of $\mathbb{R}^n$ is a real number.
- b) Using matrix notation we can write the scalar product as a matrix product of the row matrix given by $\mathbf{u}$ and the column matrix given by $\mathbf{v}$:

$$\mathbf{u} \cdot \mathbf{v} = \begin{pmatrix} u_1 & u_2 & \dots & u_n \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Example 51

Let $\mathbf{u} = (1, 2, 3)$ and $\mathbf{v} = (2, -1, 4)$, their scalar product is

$$\mathbf{u} \cdot \mathbf{v} = 1 \cdot 2 + 2 \cdot (-1) + 3 \cdot 4 = 12.$$

Observe that $\mathbf{u} \cdot \mathbf{v} = 12 = \mathbf{v} \cdot \mathbf{u}$.

Proposition. Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and c a scalar (a real number). It holds:

- a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
- b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$.
- c) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$.
- d) $\mathbf{u} \cdot \mathbf{u} \geq 0$ and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

The scalar product is used to define the concept of norm (or length) of a vector:

The **norm** or length of a vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

In $\mathbb{R}^2$ and $\mathbb{R}^3$ this concept is the usual length of the oriented segment representing a vector.

Proposition. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and c a scalar. It holds:

- a) $\|\mathbf{u}\| \geq 0$ and $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$.
- b) $\|c\mathbf{u}\| = |c|\|\mathbf{u}\|$ (for $|c|$ the absolute value of c).
- c) $\|\mathbf{u} \cdot \mathbf{v}\| \leq \|\mathbf{u}\|\|\mathbf{v}\|$ (Cauchy-Schwartz inequality).
- d) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (triangular inequality).

A **unitary vector** is a vector whose norm is 1. Observe that, given any vector $\mathbf{u}$ of $\mathbb{R}^n$ different from 0, we can always get a new unitary vector $\mathbf{v}$ with the same direction than $\mathbf{u}$ just taking $\mathbf{v} = \frac{\mathbf{u}}{\|\mathbf{u}\|}$. This process is called **normalization** of $\mathbf{u}$.

Example 52

The vector $\mathbf{w} = (0, -3/5, 4/5)$ is unitary since

$$\|\mathbf{w}\| = \sqrt{0^2 + (-3/5)^2 + (4/5)^2} = 1.$$

On the other hand, the vector $\mathbf{u} = (2, 1, 3)$ is not unitary, since

$$\|\mathbf{u}\| = \sqrt{2^2 + 1^2 + 3^2} = \sqrt{14} \neq 1.$$

We can normalize it and get $\mathbf{v} = \frac{\mathbf{u}}{\|\mathbf{u}\|} = \left(\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right)$ which is a unitary vector pointing in the same direction as $\mathbf{u}$.

The concept of norm of a vector lets us define another important concept: the distance between two points in $\mathbb{R}^n$:

Let $\mathbf{u}, \mathbf{v}$ be two vectors of $\mathbb{R}^n$, we define the distance between them as

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Note

If $\mathbf{u} = \overrightarrow{OP}$ and $\mathbf{v} = \overrightarrow{OQ}$, then $d(\mathbf{u}, \mathbf{v})$ can be interpreted as the distance between the two points P and Q , and computed as $\|\overrightarrow{PQ}\|$.

Example 53

The distance between the vectors $\mathbf{u} : (1, 2, 3)$ and $\mathbf{v} : (2, -1, 4)$ is:

$$\mathbf{u} - \mathbf{v} = (-1, 3, -1) \Rightarrow d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(-1)^2 + 3^2 + (-1)^2} = \sqrt{11}.$$

In $\mathbb{R}^2$ and $\mathbb{R}^3$ the angle between two nonzero vectors can be defined using the scalar product:

Given $\mathbf{u}, \mathbf{v}$ two vectors from $\mathbb{R}^2$ or $\mathbb{R}^3$ different than 0, we define the **angle** between them as the real number $\theta \in [0, \pi]$ such that

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Note

θ is expressed in radians

Example 54

Let's compute the angle θ between the vectors $\mathbf{u} = (1, 2, 3)$ and $\mathbf{v} = (2, -1, 4)$:

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{12}{\sqrt{14} \cdot \sqrt{21}} = 0.69985.$$

Therefore $\theta = \cos^{-1}(0.69985) = 0.7956 \text{ rad}$.

6.2. Orthogonal vectors and basis in $\mathbb{R}^n$

If $\mathbf{u} \cdot \mathbf{v} = 0$ then the angle between these two vectors is $\pi/2$, which means that the directions of both vectors are perpendicular. The following concept generalizes to $\mathbb{R}^n$ the notion of perpendicularity.

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we call them **orthogonal** (or perpendicular) whenever:

$$\mathbf{u} \cdot \mathbf{v} = 0$$

Observe that the vector $\mathbf{0} = (0, \dots, 0) \in \mathbb{R}^n$ is orthogonal to any vector of $\mathbb{R}^n$.

Example 55

Knowing that the vectors $\mathbf{v} = (2, -1, 4)$ and $\mathbf{w} = (0, 3, m)$ are orthogonal, what is the value of m ? Since they are orthogonal, we have:

$$0 = \mathbf{v} \cdot \mathbf{w} = 2 \cdot 0 + (-1) \cdot 3 + 4 \cdot m$$

therefore $m = 3/4$.

Let $\mathbf{u} \in \mathbb{R}^n$ be a vector and $W \leq \mathbb{R}^n$ be a vector subspace, we call $\mathbf{u}$ orthogonal to the whole subspace W whenever $\mathbf{u}$ is orthogonal to any vector of W .

The set of all vectors $\mathbf{u} \in \mathbb{R}^n$ orthogonal to the subspace W is called the **orthogonal complement** of W and denoted by $W^\perp$.

The orthogonal complement of a vector subspace of $\mathbb{R}^n$ is a vector subspace of $\mathbb{R}^n$ itself.

Example 56

Take the plane xy in $\mathbb{R}^3$, whose equation is $z = 0$. This plane is a vector subspace of $\mathbb{R}^3$ whose vectors have the third component equal to 0, i.e.:

$$W = \{(x, y, z) \in \mathbb{R}^3 | z = 0\} = \{(x, y, 0) | x, y \in \mathbb{R}\} = \langle (1, 0, 0), (0, 1, 0) \rangle.$$

It is obvious that $\mathbf{u} = (0, 0, 1)$ is orthogonal to W , since for any $\mathbf{w} = (w_1, w_2, 0) \in W$ we have $\mathbf{w} \cdot \mathbf{u} = w_1 \cdot 0 + w_2 \cdot 0 + 0 \cdot 1 = 0$.

In fact, if we denote $Z = \langle (0, 0, 1) \rangle$ the subspace generated by $\mathbf{u}$ (i.e., the line given by the axis z in $\mathbb{R}^3$), we have that Z is the orthogonal complement to W : $Z = W^\perp$. Observe that it is also true that $W = Z^\perp$.

Let W be a vector subspace of $\mathbb{R}^n$ and $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ a basis of W . We call B a **orthogonal basis** of W whenever the vectors of B are orthogonal two by two, i.e.:

$$\mathbf{u}_i \cdot \mathbf{u}_j = 0 \text{ for all } i \neq j.$$

On the other hand, we say that B is an **orthogonal basis** of W whenever is an orthogonal basis and all the vectors on it are unitary.

Observe that, given an orthonormal basis B it is immediate to get an orthonormal basis: we only need to normalize each vector in B . This is, let $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthogonal basis of W , then $B' = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ for $\mathbf{v}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}$ ($i = 1, \dots, k$) is an orthonormal basis of W .

Example 57

The vectors $\mathbf{u}_1 = (3, 1, 1)$, $\mathbf{u}_2 = (-1, 2, 1)$, $\mathbf{u}_3 = (-1/2, -2, 7/2)$ form a basis in $\mathbb{R}^3$ since $\langle \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \rangle = \mathbb{R}^3$ and $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a linearly independent set of vectors.

It is also an orthogonal basis, since $\mathbf{u}_1 \cdot \mathbf{u}_2 = 0$, $\mathbf{u}_2 \cdot \mathbf{u}_3 = 0$, $\mathbf{u}_1 \cdot \mathbf{u}_3 = 0$. In general, whenever a set of vectors different from zero are orthogonal two by two, they are also linearly independent.

From this orthogonal basis B we can get an orthonormal basis B' :

$$\begin{aligned} \|(3, 1, 1)\| &= \sqrt{11} \\ \|(-1, 2, 1)\| &= \sqrt{6} \\ \|(-1/2, -2, 7/2)\| &= \sqrt{33/2} \end{aligned}$$

Therefore $B' = \{(3/\sqrt{11}, 1/\sqrt{11}, 1/\sqrt{11}), (-1/\sqrt{6}, 2/\sqrt{6}, 1/\sqrt{6}), ((-1/2)/\sqrt{33/2}, -2/\sqrt{33/2}, (7/2)/\sqrt{33/2})\}$ is an orthonormal basis of $\mathbb{R}^3$.

Given a basis $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ of a subspace W of $\mathbb{R}^n$, any vector $\mathbf{v} \in W$ can be written as a linear combination of the elements in the basis, i.e.: there exist $c_1, \dots, c_k$ such that:

$$\mathbf{v} = c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k$$

i.e.:

$$(\mathbf{v}_1, \dots, \mathbf{v}_n) = c_1(u_{11}, \dots, u_{1n}) + \dots + c_k(u_{k1}, \dots, u_{kn}).$$

In general, in order to find the exact value of the k coefficients c_i we will need to solve the system of p linear equations

$$\begin{aligned} \mathbf{v}_1 &= c_1 u_{11} + c_2 u_{21} + \dots + c_k u_{k1} \\ \mathbf{v}_2 &= c_1 u_{12} + c_2 u_{22} + \dots + c_k u_{k2} \\ &\vdots \\ \mathbf{v}_n &= c_1 u_{1n} + c_2 u_{2n} + \dots + c_k u_{kn} \end{aligned}$$

However, if the basis is orthogonal, the process of finding c_i is much more simpler:

Theorem. Let $B = \{u_1, \dots, u_k\}$ an orthogonal basis of the subspace $W \leq \mathbb{R}^n$ and $\mathbf{v} \in W$. The expression of $\mathbf{v}$ as a linear combination of the elements of B is:

$$\mathbf{v} = c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k, \quad \text{for } c_i = \frac{\mathbf{v} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}, \quad i = 1, \dots, k.$$

Example 58

As previously seen, the vectors $\mathbf{u}_1 = (3, 1, 1)$, $\mathbf{u}_2 = (-1, 2, 1)$, $\mathbf{u}_3 = (-1/2, -2, 7/2)$ form an orthogonal basis of $\mathbb{R}^3$. Consider $\mathbf{v} = (6, 1, -8)$ another vector of $\mathbb{R}^3$, and let's find the expression of v in the orthogonal basis $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

If the basis B wasn't orthogonal, we should solve the vectorial equation:

$$(6, 1, -8) = c_1(3, 1, 1) + c_2(-1, 2, 1) + c_3(-1/2, -2, 7/2).$$

This vectorial equation rises to a system of three linear equations with three unknowns.

However, B is an orthogonal basis. Therefore, we can use the previous theorem to find the components of $\mathbf{v}$ in B :

$$\begin{array}{lll} \mathbf{v} \cdot \mathbf{u}_1 = 11 & \mathbf{v} \cdot \mathbf{u}_2 = -12 & \mathbf{v} \cdot \mathbf{u}_3 = -33 \\ \mathbf{u}_1 \cdot \mathbf{u}_1 = 11 & \mathbf{u}_2 \cdot \mathbf{u}_2 = 6 & \mathbf{u}_3 \cdot \mathbf{u}_3 = 33/2 \end{array}$$

Therefore:

$$c_1 = \frac{11}{11} = 1 \quad c_2 = \frac{-12}{6} = -2 \quad c_3 = \frac{-33}{33/2} = -2$$

We can conclude $\mathbf{v} = \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3$, thus the components of $\mathbf{v}$ in the basis B are $(1, -2, -2)$.

6.3. Orthogonal projections

Let $\mathbf{v} \in \mathbb{R}^n$ be a vector, $W \leq \mathbb{R}^n$ a vector subspace and $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ an orthogonal basis of W . We call **orthogonal projection** of $\mathbf{v}$ over W , $OP(\mathbf{v}, W)$, to the vector $\mathbf{v}^* \in W$:

$$OP(\mathbf{v}, W) = \mathbf{v}^* = c_1 \mathbf{u}_1 + \dots + c_k \mathbf{u}_k, \quad \text{for } c_i = \frac{\mathbf{v} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}, \quad i = 1, \dots, k$$

Example 59

Let $W = \langle \mathbf{u}_1, \mathbf{u}_2 \rangle \leq \mathbb{R}^3$ be the subspace generated by the vectors $\mathbf{u}_1 = (2, 5, -1)$, $\mathbf{u}_2 = (-2, 1, 1)$. Observe that $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis of W and W can be represented by a plane of $\mathbb{R}^3$.

We will compute the orthogonal projection of the vector $\mathbf{v} = (1, 2, 3)$ over the subspace W :

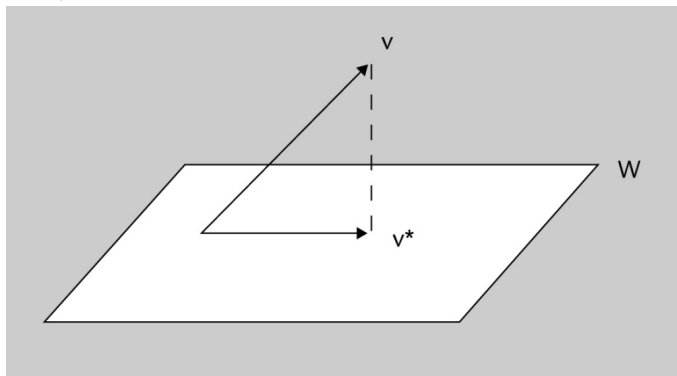
$$\begin{aligned} \mathbf{v} \cdot \mathbf{u}_1 &= 9 & \mathbf{v} \cdot \mathbf{u}_2 &= 3 \\ \mathbf{u}_1 \cdot \mathbf{u}_1 &= 30 & \mathbf{u}_2 \cdot \mathbf{u}_2 &= 6, \end{aligned}$$

Therefore:

$$c_1 = \frac{9}{30} = \frac{3}{10} \quad c_2 = \frac{3}{6} = \frac{1}{2}$$

Thus the orthogonal projection $\mathbf{v}^* = \frac{3}{10}(2, 5, -1) + \frac{1}{2}(-2, 1, 1) = (-\frac{2}{5}, 2, \frac{1}{5}) \in W$.

Figure 7. Graphical representation of the orthogonal projection of the vector $\mathbf{v} = (1, 2, 3)$ onto the subspace W .



Theorem (orthogonal decomposition). In the same conditions as the previous definition, any vector $\mathbf{v} \in \mathbb{R}^n$ can be written in such a way

$$\mathbf{v} = \mathbf{v}^* + \mathbf{z}$$

that $\mathbf{v}^*$ is the orthogonal projection of $\mathbf{v}$ over W and $\mathbf{z} = \mathbf{v} - \mathbf{v}^*$ a vector of $W^\perp$.

Example 60

Following the previous example, let's compute the orthogonal decomposition of $\mathbf{v}$ as the sum of its orthogonal projection $\mathbf{v}^*$ over W and a new vector $\mathbf{z}$ orthogonal to W .

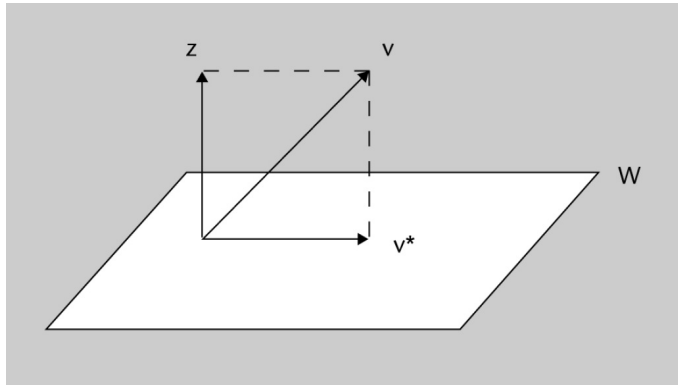
As we have already seen $\mathbf{v}^* = (-\frac{2}{5}, 2, \frac{1}{5}) \in W$.

On the other hand $\mathbf{z} = \mathbf{v} - \mathbf{v}^* = (1, 2, 3) - (-\frac{2}{5}, 2, \frac{1}{5}) = (\frac{7}{5}, 0, \frac{14}{5})$.

In order to prove that $\mathbf{z} \in W^\perp$ we only have to check that $\mathbf{z}$ is orthogonal to the vectors of a basis of W :

$$\mathbf{z} \cdot \mathbf{u}_1 = 2 \cdot \frac{7}{5} + 5 \cdot 0 + (-1) \cdot \frac{14}{5} = 0$$

$$\mathbf{z} \cdot \mathbf{u}_2 = (-2) \cdot \frac{7}{5} + 1 \cdot 0 + 1 \cdot \frac{14}{5} = 0$$

Figure 8. Orthogonal decomposition of the vector $\mathbf{v}$ 

Observe that, if $\mathbf{v} = \mathbf{v}^* + \mathbf{z}$, then $\mathbf{z} = \mathbf{v} - \mathbf{v}^*$ is the component of $\mathbf{v}$ orthogonal to W . This idea is the key point to understanding the process of orthogonalization of Gram-Schmidt that we will introduce in the next section.

Theorem (optimal approximation). In the same conditions as the previous definition, the vector $\mathbf{v}^*$ is the nearest vector of W to $\mathbf{v}$, i.e.: $d(\mathbf{v}, \mathbf{v}^*) < d(\mathbf{v}, \mathbf{u})$ for any other vector $\mathbf{u} \in W$ different from $\mathbf{v}^*$.

The distance between a vector $\mathbf{v} \in \mathbb{R}^n$ to the subspace $W \leq \mathbb{R}^n$ is defined as the distance from $\mathbf{v}$ to the nearest vector of W , this is, the distance from $\mathbf{v}$ to $\mathbf{v}^*$.

Example 61

Let $\mathbf{u}_1 = (5, -2, 1)$ and $\mathbf{u}_2 = (1, 2, -1)$. What is the distance between the vector $\mathbf{v} = (-1, -5, 10)$ and the subspace $W = \langle \mathbf{u}_1, \mathbf{u}_2 \rangle$? Using the theorem of optimal approximation, the distance between $\mathbf{v}$ and W is the same as the distance between $\mathbf{v}$ and $\mathbf{v}^*$, the projection of $\mathbf{v}$ over W .

We will check first that the basis $B = \{\mathbf{u}_1, \mathbf{u}_2\}$ is orthogonal:

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = 5 \cdot 1 + (-2) \cdot 2 + 1 \cdot (-1) = 0.$$

Now we look for $\mathbf{v}^*$:

$$\mathbf{v} \cdot \mathbf{u}_1 = 15, \quad \mathbf{u}_1 \cdot \mathbf{u}_1 = 30; \quad \mathbf{v} \cdot \mathbf{u}_2 = -21, \quad \mathbf{u}_2 \cdot \mathbf{u}_2 = 6;$$

so that:

$$c_1 = \frac{15}{30} = \frac{1}{2} \quad c_2 = \frac{-21}{6} = -\frac{7}{2}$$

$$\text{Therefore } \mathbf{v}^* = \frac{1}{2}(5, -2, 1) + \frac{-7}{2}(1, 2, -1) = (-1, -8, 4) \in W.$$

$$\text{Finally: } d(\mathbf{v}, W) = d(\mathbf{v}, \mathbf{v}^*) = \|\mathbf{v} - \mathbf{v}^*\| = \|(0, 3, 6)\| = 3\sqrt{5}.$$

6.4. Gram-Schmidt orthogonalization method

Sometimes it can be pretty useful to have an orthonormal basis of a vector subspace $W \leq \mathbb{R}^n$. The Gram-Schmidt method is an algorithm that gives an orthogonal basis for any non-trivial subspace of $\mathbb{R}^n$ (a trivial subspace is one that only has the vector

$\mathbf{0} \in \mathbb{R}^n$). Once we already have an orthogonal basis, getting an orthonormal basis is trivial since we only have to normalize the vectors of such orthogonal basis.

Gram-Schmidt orthogonalization method. Let $W \leq \mathbb{R}^n$ be a subspace and $B = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ a basis of W .

- 1: Take $\mathbf{v}_1 = \mathbf{u}_1$ and consider $W_1 = \langle \mathbf{v}_1 \rangle$.
- 2: Take $\mathbf{v}_2 = \mathbf{u}_2 - OP(\mathbf{u}_2, W_1)$ and consider $W_2 = \langle \mathbf{v}_1, \mathbf{v}_2 \rangle$.
- 3: Take $\mathbf{v}_3 = \mathbf{u}_3 - OP(\mathbf{u}_3, W_2)$ and consider $W_3 = \langle \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \rangle$.
- 4: Take $\mathbf{v}_4 = \mathbf{u}_4 - OP(\mathbf{u}_4, W_3)$ and consider $W_4 = \langle \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \rangle$.
- $\vdots$
- k : Take $\mathbf{v}_k = \mathbf{u}_k - OP(\mathbf{u}_k, W_{k-1})$.

The basis $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is orthogonal.

Example 62

Let $\mathbf{u}_1 = (1, 1, 1, 1)$, $\mathbf{u}_2 = (0, 1, 1, 1)$, $\mathbf{u}_3 = (0, 0, 1, 1)$ be three vectors of $\mathbb{R}^4$. The vectors $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ are linearly independent and they form a basis of a subspace W of dimension 3.

Let's use the Gram-Schmidt method to find an orthogonal basis of W :

- 1) Let $\mathbf{v}_1 = (1, 1, 1, 1)$ and $W_1 = \langle \mathbf{v}_1 \rangle$.
- 2) Let $\mathbf{v}_2 = (0, 1, 1, 1) - OP(\mathbf{u}_2, W_1)$. Computing we get:

$$\begin{aligned}\mathbf{u}_2 \cdot \mathbf{v}_1 &= 3 \\ \mathbf{v}_1 \cdot \mathbf{v}_1 &= 4 \\ c_1 &= \frac{3}{4}.\end{aligned}$$

Therefore $\mathbf{v}_2 = (0, 1, 1, 1) - \frac{3}{4}(1, 1, 1, 1) = (-\frac{3}{4}, 1/4, 1/4, 1/4)$.

We can just take the vector $\mathbf{v}_2$ obtained and continue with the algorithm. However, in order to simplify the computations, we will take instead the proportional vector $\mathbf{v}'_2 = 4\mathbf{v}_2 = (-3, 1, 1, 1)$.

We define $W_2 = \langle \mathbf{v}_1, \mathbf{v}'_2 \rangle$. Observe that $\mathbf{v}_1, \mathbf{v}'_2$ are orthogonal vectors since $\mathbf{v}_2 \in W_1^\perp$ (because of the orthogonal decomposition theorem), therefore $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis of W_2 .

- 3) Let $\mathbf{v}_3 = (0, 0, 1, 1) - OP(\mathbf{u}_3, W_2)$. Computing we get:

$$\begin{aligned}\mathbf{u}_3 \cdot \mathbf{v}_1 &= 2 & \mathbf{u}_3 \cdot \mathbf{v}'_2 &= 2 \\ \mathbf{v}_1 \cdot \mathbf{v}_1 &= 4 & \mathbf{v}'_2 \cdot \mathbf{v}'_2 &= 12 \\ c_1 &= \frac{2}{4} = \frac{1}{2} & c_2 &= \frac{2}{12} = \frac{1}{6}\end{aligned}$$

Therefore $\mathbf{v}_3 = (0, 0, 1, 1) - \frac{1}{2}(1, 1, 1, 1) - \frac{1}{6}(-3, 1, 1, 1) = (0, -2/3, 1/3, 1/3)$.

In order to simplify the computations, we will take $\mathbf{v}'_3 = (0, -2, 1, 1)$.

Observe that $\mathbf{v}'_3$ is orthogonal to $\mathbf{v}_1$ and $\mathbf{v}'_2$ since $\mathbf{v}_3 \in W_2^\perp$ (because of the orthogonal decomposition theorem), therefore $B = \{\mathbf{v}_1, \mathbf{v}'_2, \mathbf{v}'_3\}$ is an orthogonal basis of W_3 .

Following the previous theorem, B is also an orthogonal basis of W . In order to get an orthonormal basis we just have to normalize the vectors of this basis:

$$\|\mathbf{v}_1\| = 2 \quad \|\mathbf{v}_2'\| = 2\sqrt{3} \quad \|\mathbf{v}_3'\| = \sqrt{6}$$

Therefore $B' = \left\{ \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right), \left(\frac{-3}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}} \right), \left(0, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) \right\}$ is an orthonormal basis of W .

Summary

In this module we reviewed methods and concepts associated to linear algebra and geometry, which are basic for the development of the following modules (linear systems of equations, geometric transformations, linear applications, etc.).

First, we reviewed the following key concepts associated to real vector spaces:

- vector subspace
- linear combination of vectors
- system of generators
- linear independence
- rank of a set of vectors
- basis and dimension of a vector space

Next, we reviewed the key concepts associated to the theory of matrices and determinants:

- dimension of a matrix
- transposed matrix
- types of matrices (diagonal, symmetric, triangular, etc.)
- operations with matrices (addition, subtraction, product, etc.)
- inverse matrix
- determinants of order 2 and 3
- computation of determinants by adjoints
- properties of determinants
- computation of an inverse matrix
- computation of the rank of a matrix
- determinants and linear combinations

In the final part of the module, we reviewed the following concepts on geometry:

- equations of lines and planes in 2D and 3D
- scalar product of $\mathbb{R}^n$ and angle between two vectors
- orthogonality (orthogonal and orthonormal basis)
- Gram-Schmidt orthogonalization method

The module has been completed with examples and solved activities, and some interesting applications of the theory to different disciplines (computer sciences, telecommunications, economy...) have been introduced as well.

Self-assessment exercises

1. Find if the following sets of vectors are a basis of $\mathbb{R}^3$:

- a) $\{(-2, 3, 0), (3, -1, 2), (-1, 5, 2)\}$
- b) $\{(-1, 2, 1), (2, 4, 0), (5, 1, 1)\}$

2. Compute the inverse of the following matrix using determinants:

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -3 \\ 0 & 1 & 1 \\ 3 & 1 & -2 \end{pmatrix}$$

3. Let $\mathbf{A} = \begin{pmatrix} 5 & -4 & 2 \\ 2 & -1 & 1 \\ -4 & 4 & -1 \end{pmatrix}$, compute $\mathbf{A}^2 - 2\mathbf{A}$.

To do with or without software.

4. Consider the following vectors: $\mathbf{u} = (a, 1, -2)$, $\mathbf{v} = (1, a, 2)$, $\mathbf{w} = (2a, 1, 0)$. Find the value of a such that the three vectors are linearly independent.

5. Given the equation $\mathbf{X} \cdot \mathbf{A} = \mathbf{B}$, find the matrix $\mathbf{X}$ for $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 3 & 1 & 2 \end{pmatrix}$ and

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

6. Compute the inverse matrix, in terms of the parameter a , of $\begin{pmatrix} a & 0 & 0 \\ 1 & a & 2 \\ 1 & 2 & a \end{pmatrix}$.

To do with or without software.

7. Find the dimension of the generated subspace by the vectors $(1, 2, 0)$, $(1, 0, 1)$ and $(-2, 2, -3)$. Compute k such that the vector $(4, 3, k)$ belongs to such subspace.

8. Given the sets of vectors $B = \{(1, 1, 2), (1, 2, 3), (3, 4, 3)\}$ and $A = \{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$:

- a) Check if B and A are basis of $\mathbb{R}^3$ and compute the components of the vector $\mathbf{v} = (1, 1, 1)$ in B and in A , respectively.
- b) Compute the change-of-basis matrix from B to A , and check the results found previously.

9. Discuss if the following sentences on the scalar product of vectors of $\mathbb{R}^n$ are true or false:

- a) The norm of a vector is a positive number.
- b) The distance between $\mathbf{u}$ and $\mathbf{v}$ is $\|\mathbf{u} - \mathbf{v}\|$.
- c) Two vectors different from 0 are orthogonal if and only if they are linearly independent.
- d) The orthogonal projection of $\mathbf{u}$ over $\mathbf{v}$ is a multiple of $\mathbf{u}$.
- e) If a vector coincides with its orthogonal projection over a subspace, then the vector belongs to such subspace.
- f) The subspace of all the vectors of $\mathbb{R}^n$ orthogonal to a certain vector is a subspace of $\mathbb{R}^n$.

10. Let $W = \{(x, y, z) \in \mathbb{R}^3 | x + y + z = 0\}$ be a vector subspace and $\mathbf{v} = (2, -1, -1)$.

- a) Check that $B_1 = \{(-1, 0, 1), (-1, 3, -2)\}$ is a basis of W .
- b) Compute the components of $\mathbf{v}$ in the basis B_1 .
- c) Taking $B_2 = \{(-1, 0, 1), (0, 1, -1)\}$ another basis of W , compute the components of $\mathbf{v}$ in the basis B_2 .
- d) Compute the change-of-basis matrix from B_1 to B_2 and check the coherence of the components of $\mathbf{v}$ in both basis.

11. A Leontief model of production is expressed through an *input-output* table such as:

Table 2. Leontief production model

	Sector I	Sector II	Sector III
Sector I	0.5	0.4	0.2
Sector II	0.2	0.3	0.1
Sector III	0.1	0.1	0.3

To do with software.

This table is read in the following way: if the sector II wants to produce 1000 units, then it will need 400 units from sector I, 300 from sector II and 100 from sector III:

Table 3. Interpretation of Leontief production model

	Sector I	Sector II	Sector III
Sector I	0.5	$0.4 \times 1000 = 400$ units of sector I	0.2
Sector II	0.2	$0.3 \times 1000 = 300$ units of sector II	0.1
Sector III	0.1	$0.1 \times 1000 = 100$ units of sector III	0.3

The matrix $\mathbf{C}$ represents the consumption of an economy, which in this case would be:

$$\mathbf{C} = \begin{pmatrix} 0.5 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.1 \\ 0.1 & 0.1 & 0.3 \end{pmatrix}$$

If the total demand of consumers is given by the vector (d_1, d_2, d_3) , where d_i is the demand from sector $i = 1, 2, 3$, then the quantity produced in the economy (x_1, x_2, x_3) satisfies the following equation:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \mathbf{C} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

- What is the required level of production (x_1, x_2, x_3) to cover a demand of $d_1 = 50$, $d_2 = 30$, $d_3 = 20$?
- If the level of production is $x_1 = 4350$, $x_2 = 3480$, $x_3 = 1958$, what level of demand can it cover?

12. Compute, using the properties of the determinant, the determinant associated to the

matrix $\mathbf{A} = \begin{pmatrix} 1 & a & b+c \\ 1 & b & a+c \\ 1 & c & a+b \end{pmatrix}$.

13. Let $P : (-1, 0, 3)$ be a point and $\mathbf{v} = (2, -1, 4)$ be a vector. Find:

- All the equations of the line l in the direction $\mathbf{v}$ passing through P .
- The missing coordinates from $Q : (x, y, -3)$ such that it belongs to l .

14. Let $P : (-2, 1, 1)$ be a point, and $\mathbf{u} = (3, -1, 2)$ and $\mathbf{v} = (2, -2, 1)$ be two vectors. Find:

- All the equations of the plane π in the directions $\mathbf{u}, \mathbf{v}$ passing through P .
- The missing coordinates from $Q : (x, -3, 0)$ such that it belongs to π .

To do with software.

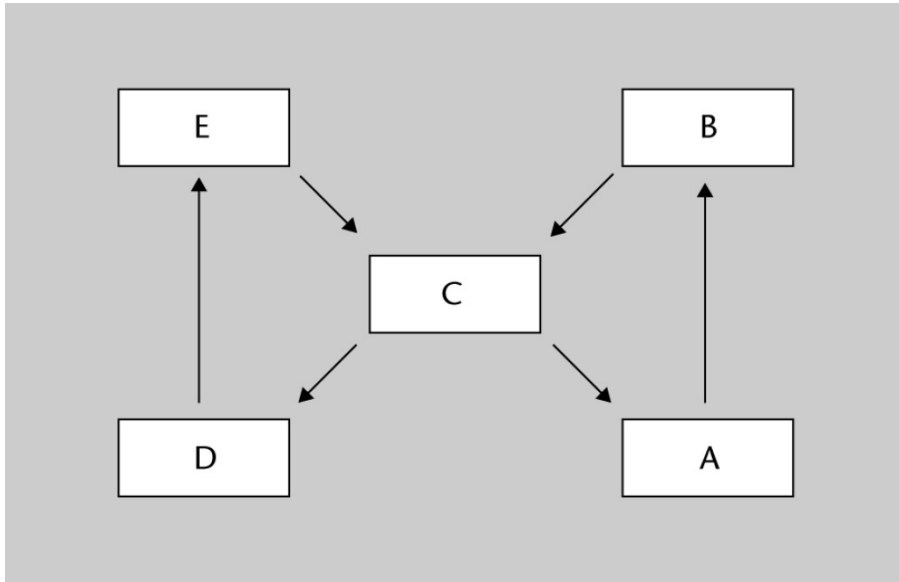
15. The graph in figure 9, in the next page, shows the direct link between five websites belonging to different companies on the sector of simulation software development:

- Represent with a matrix $\mathbf{M}$, whose elements are 0 and 1, the information presented on direct links in the graph, i.e.: the element $(\mathbf{M})_{ij}$ is 1 if there exist a direct link between the company represented on row i and the company represented on column j (for $i \neq j$), or 0 in case it doesn't exist.

To do with software.

- b) Represent with a matrix the information given in the graph on non-direct links with a single website of separation between them (e.g., there is a link of this kind between the sites A and C , since we can get from A to C passing through B).
- c) Compute $\mathbf{M}^2$. What do you observe?
- d) Compute $\mathbf{M} + \mathbf{M}^2$ and explain the result.

Figure 9. Graph representing the direct links between five websites



16. An easy way of coding a text message is to assign every letter to a single number. For example, assigning $A = 01$, $B = 02$, ..., $Z = 27$ and the blank space 28, the message “tomorrow day D” codifies as (ignoring capital letters) “20 15 13 15 18 18 15 23 28 04 01 25 28 04”. The previous succession can be written as a matrix, completing it with blank spaces (value 28) if necessary.

- a) Write a matrix representing the previous succession. Why do you believe that you can decode easily the original text with it?
- b) Multiply the previous matrix by a second matrix (code-decode matrix that, in general, cannot have less rows than columns and must have maximum rank). Do you think that it will be more difficult to obtain the original text from the resulting matrix now (without any knowledge about the code-decode matrix)?

To do with software

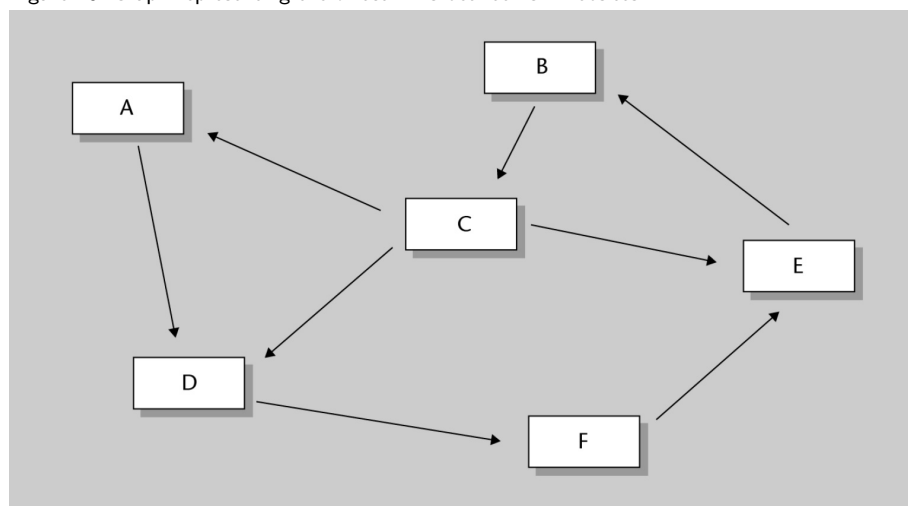
17. Given a sets of vectors $B = \{(0, 1, -2), (5, -7, 4), (6, 3, 5)\}$ and $A = \{(1, 1, -2), (-5, -1, 2), (7, 0, -5)\}$ of $\mathbb{R}^3$:

- a) Check that B and A are two basis of $\mathbb{R}^3$ and compute the components of the vector $\mathbf{v} = (2, 1, -1)$ in both basis.
- b) Compute the change-of-basis matrix from B to A and check the coherence of the previous result.

18. The graph in figure 10, in the next page, shows the direct link between five websites belonging to different companies on the sector of simulation software development:

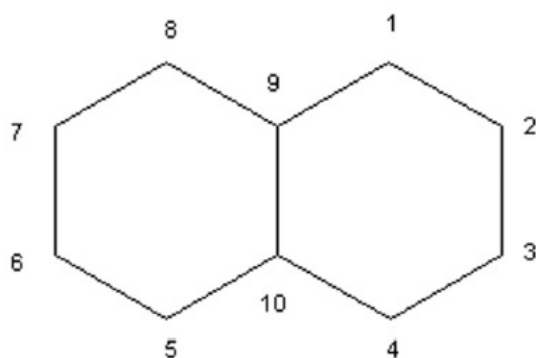
- a) Represent with a matrix $\mathbf{M}$ whose elements are 0 and 1 the information presented on direct links in the graph, i.e.: the element $(\mathbf{M})_{ij}$ is 1 if there exist a direct link between the company represented on row i and the company represented on column j (for $i \neq j$), or 0 in case it doesn't exist.
- b) Using the matrix $\mathbf{M}$ find the matrix $\mathbf{N}$ representing the links until two connections (i.e. direct links or those with one or two connections) between the websites. Give an interpretation of such matrix $\mathbf{N}$.

Figure 10. Graph representing the direct links between six websites



19. Consider the graph associated to the naphthalene molecule:

Figure 11. Naphthalene molecule



- Represent with a matrix $\mathbf{M}$ whose elements are 0 and 1 the information presented on the previous graph.
- What is the dimension of this matrix?
- Compute the transposed matrix, the inverse matrix and its determinant.
- What can you say about $\mathbf{M} \cdot \mathbf{M}^{-1}$? And about $\mathbf{M}^{-1} \cdot \mathbf{M}$? After the results, can you state that the matrix product is commutative?

Solutions

Introductory example

Remember that the table of state change is:

Table 1. Probabilities that a LAN net will go from state X to state Y

	Today (day n)		
Tomorrow (day $n + 1$)	State A	State B	State C
State A	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
State B	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$
State C	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$

We can represent, respectively, the matrix of state change and the vector of initial estate as:

$$\mathbf{M} = \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{8} & \frac{1}{4} & \frac{1}{4} \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \begin{array}{l} \leftarrow P(A_0) \\ \leftarrow P(B_0) \\ \leftarrow P(C_0) \end{array}$$

We will use the following notation:

$$\begin{aligned} A_k &= \text{“In } k \text{ days, the LAN will be in state A”} \\ B_k &= \text{“In } k \text{ days, the LAN will be in state B”} \\ C_k &= \text{“In } k \text{ days, the LAN will be in state C”} \end{aligned}$$

By probability theory, we know that

$$\begin{aligned} P(A_1) &= P([A_0 \cap A_1] \cup [B_0 \cap A_1] \cup [C_0 \cap A_1]) \\ &= P([A_0 \cap A_1]) + P([B_0 \cap A_1]) + P([C_0 \cap A_1]) \\ &= P(A_1|A_0) \cdot P(A_0) + P(A_1|B_0) \cdot P(B_0) + P(A_1|C_0) \cdot P(C_0) \\ &= M_{11} \cdot v_1 + M_{12} \cdot v_2 + M_{13} \cdot v_3 \end{aligned}$$

In a similar way, it also holds:

$$\begin{aligned} P(B_1) &= M_{21} \cdot v_1 + M_{22} \cdot v_2 + M_{23} \cdot v_3 \\ P(C_1) &= M_{31} \cdot v_1 + M_{32} \cdot v_2 + M_{33} \cdot v_3 \end{aligned}$$

In other words, if we denote $\mathbf{X}_k$ to the vector with the probabilities that the LAN is in state A, B or C after k days, we have:

$$\mathbf{X}_1 = \begin{pmatrix} P(A_1) \\ P(B_1) \\ P(C_1) \end{pmatrix} = \mathbf{M} \cdot \mathbf{v} = \begin{pmatrix} 0.375 \\ 0.375 \\ 0.25 \end{pmatrix}$$

In general, it can be proved easily that $\mathbf{X}_k = \mathbf{M}^k \cdot \mathbf{v}$, which lets us answer the questions for the first block:

$$\mathbf{X}_2 = \mathbf{M}^2 \cdot \mathbf{v} \implies \mathbf{X}_2 = \begin{pmatrix} 0.531 \\ 0.266 \\ 0.203 \end{pmatrix} \begin{array}{l} \leftarrow P(A_2) \\ \leftarrow P(B_2) \\ \leftarrow P(C_2) \end{array}$$

$$\mathbf{X}_7 = \mathbf{M}^7 \cdot \mathbf{v} \implies \mathbf{X}_7 = \begin{pmatrix} 0.608 \\ 0.218 \\ 0.174 \end{pmatrix} \begin{array}{l} \leftarrow P(A_7) \\ \leftarrow P(B_7) \\ \leftarrow P(C_7) \end{array}$$

Therefore, the probability that the LAN is working perfectly after one, two and seven days is, respectively, 0.375, 0.531 and 0.608.

Self-assessment exercises

1.

a) The set of vectors $\{(-2, 3, 0), (3, -1, 2), (-1, 5, 2)\}$ is linearly dependent, since

$$\begin{vmatrix} -2 & 3 & 0 \\ 3 & -1 & 2 \\ -1 & 5 & 2 \end{vmatrix} = 0$$

Therefore, it cannot be a basis of $\mathbb{R}^3$.

b) The set $\{(-1, 2, 1), (2, 4, 0), (5, 1, 1)\}$ are linearly independent, since

$$\begin{vmatrix} -1 & 2 & 1 \\ 2 & 4 & 0 \\ 5 & 1 & 1 \end{vmatrix} = -26 \neq 0$$

Therefore it is a basis of $\mathbb{R}^3$.

2.

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}^t$$

for $\mathbf{A}_{ij}$ is the adjoint of the element a_{ij} in the matrix $\mathbf{A}$.

Since $\det(\mathbf{A}) = 6 \neq 0$ the inverse matrix exists.

$$\text{adj}(\mathbf{A}) = \begin{pmatrix} -3 & 3 & -3 \\ -1 & 5 & 1 \\ 4 & -2 & 2 \end{pmatrix} \text{ so that}$$

$$\mathbf{A}^{-1} = \frac{1}{6} \begin{pmatrix} -3 & 3 & -3 \\ -1 & 5 & 1 \\ 4 & -2 & 2 \end{pmatrix}^t = \frac{1}{6} \begin{pmatrix} -3 & -1 & 4 \\ 3 & 5 & -2 \\ -3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{-3}{6} & \frac{-1}{6} & \frac{4}{6} \\ \frac{3}{6} & \frac{5}{6} & \frac{-2}{6} \\ \frac{-3}{6} & \frac{1}{6} & \frac{2}{6} \end{pmatrix} = \begin{pmatrix} \frac{-1}{2} & \frac{-1}{6} & \frac{2}{3} \\ \frac{1}{2} & \frac{5}{6} & \frac{-1}{3} \\ \frac{-1}{2} & \frac{1}{6} & \frac{1}{3} \end{pmatrix}$$

3.

$$\mathbf{A}^2 = \begin{pmatrix} 9 & -8 & 4 \\ 4 & -3 & 2 \\ -8 & 8 & -3 \end{pmatrix}$$

$$\mathbf{A}^2 - 2\mathbf{A} = \begin{pmatrix} 9 & -8 & 4 \\ 4 & -3 & 2 \\ -8 & 8 & -3 \end{pmatrix} - 2 \begin{pmatrix} 5 & -4 & 2 \\ 2 & -1 & 1 \\ -4 & 4 & -1 \end{pmatrix} = \begin{pmatrix} 9-10 & -8+8 & 4-4 \\ 4-4 & -3+2 & 2-2 \\ -8+8 & 8-8 & -3+2 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = -\mathbf{I}_3$$

4. For the vectors to be linearly independent we need their determinant different from 0.

Let $\mathbf{A} = \begin{pmatrix} a & 1 & 2a \\ 1 & a & 1 \\ -2 & 2 & 0 \end{pmatrix}$, whose determinant is $\det(\mathbf{A}) = 2a - 2 + 4a^2$.

This determinant is 0 for $a = 1/2$ and $a = -1$. Therefore, for any value of a different from those two, the vectors are linearly independent, thus a basis of $\mathbb{R}^3$ (because they are three vectors).

5. First we have to check that $\mathbf{A}$ is a regular matrix (invertible). For that, we compute its determinant and check that it's different from 0:

$$\begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 3 & 1 & 2 \end{vmatrix} = 2 - 3 = -1 \neq 0$$

First we need to isolate $\mathbf{X}$ in the matrix expression, and for that we need to multiply by $\mathbf{A}^{-1}$ both parts of the equality **on the right**, so that:

$$\mathbf{X} \cdot \mathbf{A} \cdot \mathbf{A}^{-1} = \mathbf{B} \cdot \mathbf{A}^{-1} \Rightarrow \mathbf{X} = \mathbf{B} \cdot \mathbf{A}^{-1}$$

Now we only need to compute the inverse of $\mathbf{A}$ and the matrix product $\mathbf{B} \cdot \mathbf{A}^{-1}$:

$$\mathbf{X} = \mathbf{B} \cdot \mathbf{A}^{-1} = \begin{pmatrix} -5 & 10 & 2 \\ 0 & 1 & 0 \\ 2 & -3 & 1 \end{pmatrix}$$

6. The inverse exists whenever the determinant of the matrix is different from 0:

$$\begin{vmatrix} a & 0 & 0 \\ 1 & a & 2 \\ 1 & 2 & a \end{vmatrix} = a(a^2 - 4) = 0 \Leftrightarrow a = 0 \text{ or } a = \pm 2$$

Therefore the inverse exists whenever $a \neq 0, 2, -2$.

We construct the matrix with the adjoints:

$$\begin{pmatrix} a^2 - 4 & 2 - a & 2 - a \\ 0 & a^2 & -2a \\ 0 & -2a & a^2 \end{pmatrix}$$

We transpose it:

$$\begin{pmatrix} a^2 - 4 & 0 & 0 \\ 2 - a & a^2 & -2a \\ 2 - a & -2a & a^2 \end{pmatrix}$$

And we divide it by the determinant of the matrix:

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{a^2-4} & 0 & 0 \\ \frac{-1}{a(2+a)} & \frac{a}{a^2-4} & \frac{-2}{a^2-4} \\ \frac{-1}{a(2+a)} & \frac{-2}{a^2-4} & \frac{a}{a^2-4} \end{pmatrix}$$

7. The dimension of the generated subspace is the rank of the matrix constructed by the generating vectors:

$$\mathbf{rk} \begin{pmatrix} 1 & 1 & -2 \\ 2 & 0 & 2 \\ 0 & 1 & -3 \end{pmatrix} = 2$$

Therefore the dimension of the subspace is 2.

Since the minor $\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} \neq 0$ the vectors that give this minor, $(1, 2, 0)$ and $(1, 0, 1)$, are linearly independent and then form a basis of the subspace.

If we want $(4, 3, k) \in \langle (1, 2, 0), (1, 0, 1) \rangle$ this vector must be linearly dependent to the vectors forming the basis of the subspace, this is:

$$\begin{vmatrix} 1 & 1 & 4 \\ 2 & 0 & 3 \\ 0 & 1 & k \end{vmatrix} = 0 \Leftrightarrow 5 - 2k = 0$$

Therefore $k = 5/2$.

8.

a) $\begin{vmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 2 & 3 & 3 \end{vmatrix} \neq 0$ and $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} \neq 0$ therefore both sets of vectors are linearly independent, and since $\dim(\mathbb{R}^3) = 3$ both sets of vectors are basis of $\mathbb{R}^3$.

The components of $\mathbf{v}$ in B are:

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 2 & 3 & 3 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{4} \\ \frac{1}{4} \end{pmatrix}$$

The components of $\mathbf{v}$ in A are:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

Those computations can be checked using software.

- b) The change-of-basis matrix from B to A is:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 2 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix}$$

Finally, we check using this matrix the results previously found:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{4} \\ \frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

9.

- a) True
- b) True
- c) True
- d) False
- e) True
- f) True

10.

- a) Let's check first $B_1 \subset W$. We only have to check that both vectors satisfy the defining equation of W : $x+y+z=0$

$$\begin{aligned} -1 + 0 + 1 &= 0 \Rightarrow (-1, 0, 1) \in W \\ -1 + 3 + (-2) &= 0 \Rightarrow (-1, 3, -2) \in W \end{aligned}$$

We also check if the matrix consisting on the two vectors has maximum rank:

$$\begin{pmatrix} -1 & 0 & 1 \\ -1 & 3 & -2 \end{pmatrix}$$

Indeed:

$$\begin{vmatrix} -1 & 0 \\ -1 & 3 \end{vmatrix} = -3 \neq 0$$

Therefore both vectors are linearly independent.

Furthermore, we have to prove that it is a generating system of W , that is, that it can generate any value of the subspace. First, we build the system resulting from the linear combination of the vectors of B_1 , with three equations and two unknowns.

$$\begin{aligned} \{(x, y, z) \in \mathbb{R}^3 | x + y + z = 0\} &\rightarrow (x, y, -x - y) = \alpha(-1, 0, 1) + \beta(-1, 3, -2) \\ &= (-\alpha - \beta, 3\beta, \alpha - 2\beta). \end{aligned}$$

Then, we observe that it is a consistent and independent system, that is, there exist (α, β) as a solution; this solution will be unique and different for each vector $(x, y, z) \in W$. To verify this, we compute the rank of the coefficient matrix $\mathbf{A}$ and the augmented matrix of the system $\mathbf{M}$:

$$\mathbf{A} = \begin{pmatrix} -\alpha & -\beta \\ 0 & 3\beta \\ \alpha & -2\beta \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{cc|c} -\alpha & -\beta & x \\ 0 & 3\beta & y \\ \alpha & -2\beta & -x - y \end{array} \right).$$

On the one hand, we have that the range of $\mathbf{A}$ is 2, since we can easily find a non-zero minor of order 2; on the other hand, the determinant $\det(\mathbf{M}) = 0$ and, therefore, its rank will also be 2. In conclusion, $\text{rg}(\mathbf{A}) = \text{rg}(\mathbf{M}) = n = 2$, so the system is consistent and independent.

b) We will check first if $(2, -1, -1) \in W$:

$$2 + (-1) + (-1) = 0 \Rightarrow (2, -1, -1) \in W.$$

Now, the vector $\mathbf{v}$ in the basis is written as (a, b) for:

$$(2, -1, -1) = a(-1, 0, 1) + b(-1, 3, -2)$$

which is the system of three linear equations:

$$\begin{cases} 2 &= -a - b \\ -1 &= 3b \\ -1 &= a - 2b \end{cases} \Rightarrow a = -\frac{5}{3}, b = -\frac{1}{3}$$

Therefore, $\mathbf{v} = (-\frac{5}{3}, -\frac{1}{3})$ in this basis.

c) Now, the vector $\mathbf{v}$ in the basis is written as (a, b) for:

$$(2, -1, -1) = a(-1, 0, 1) + b(0, 1, -1)$$

which is the system of three linear equations:

$$\begin{cases} 2 &= -a \\ -1 &= b \\ -1 &= a - b \end{cases} \Rightarrow a = -2, b = -1$$

Therefore, $\mathbf{v} = (-2, -1)$ in this basis.

- d) We write the vectors of B_1 as a linear combination of the vectors of B_2 :

$$\begin{aligned} (-1, 0, 1) &= a(-1, 0, 1) + b(0, 1, -1) \Rightarrow a = 1 \text{ and } b = 0 \\ (-1, 3, -2) &= a(-1, 0, 1) + b(0, 1, -1) \Rightarrow a = 1 \text{ and } b = 3 \end{aligned}$$

So that we can write the vectors $(-1, 0, 1)$ as $(1, 0)$ and $(-1, 3, -2)$ as $(1, 3)$ in the basis B_2 . Then, the change-of-basis matrix from B_1 to B_2 is the matrix after writing those vectors as columns:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$$

Indeed, if we multiply the vector $\mathbf{v}$ written in basis B_2 , the result is the vector $\mathbf{v}$ written in the basis B_2 :

$$\mathbf{A} \cdot \mathbf{v} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} -\frac{5}{3} \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

11.

- a) From the equation we can isolate the production value $\mathbf{X}$ as follows:

$$\mathbf{X} = \mathbf{C}\mathbf{X} + \mathbf{D} \Rightarrow \mathbf{I}\mathbf{X} - \mathbf{C}\mathbf{X} = \mathbf{D} \Rightarrow \mathbf{X} = (\mathbf{I} - \mathbf{C})^{-1}\mathbf{D}$$

therefore we can find the demand from the inverse of the matrix $\mathbf{I} - \mathbf{C}$. The resulting production vector is given by $x_1 = 225.9$, $x_2 = 118.5$ and $x_3 = 77.8$.

- b) In order to find the demand that can be satisfied from the produced quantities, we only have to multiply $(\mathbf{I} - \mathbf{C}) \cdot \mathbf{X} = \mathbf{D}$.

Since

$$\mathbf{I} - \mathbf{C} = \begin{pmatrix} 0.5 & -0.4 & -0.2 \\ -0.2 & 0.7 & -0.1 \\ -0.1 & -0.1 & 0.7 \end{pmatrix}$$

The demand that can be satisfied is:

$$\mathbf{X} = \begin{pmatrix} 4350 \\ 3480 \\ 1958 \end{pmatrix} \quad \mathbf{D} = (\mathbf{I} - \mathbf{C}) \cdot \mathbf{X} = \begin{pmatrix} 391.4 \\ 1370.2 \\ 587.6 \end{pmatrix}$$

12.

$$\begin{pmatrix} 1 & a & b+c \\ 1 & b & a+c \\ 1 & c & a+b \end{pmatrix} = \begin{pmatrix} 1 & a & a+b+c \\ 1 & b & b+a+c \\ 1 & c & c+a+b \end{pmatrix} = (a+b+c) \begin{pmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{pmatrix} = 0.$$

Or also

$$\begin{pmatrix} 1 & a & b+c \\ 1 & b & a+c \\ 1 & c & a+b \end{pmatrix} = \begin{pmatrix} 1 & a & b \\ 1 & b & c \\ 1 & c & a \end{pmatrix} + \begin{pmatrix} 1 & a & c \\ 1 & b & a \\ 1 & c & b \end{pmatrix} = \begin{pmatrix} 1 & a & b \\ 1 & b & c \\ 1 & c & a \end{pmatrix} - \begin{pmatrix} 1 & c & a \\ 1 & a & b \\ 1 & b & c \end{pmatrix} = \begin{pmatrix} 1 & a & b \\ 1 & b & c \\ 1 & c & a \end{pmatrix} - \begin{pmatrix} 1 & a & b \\ 1 & b & c \\ 1 & c & a \end{pmatrix} = 0$$

13.

- a) Vectorial equation: $(x, y, z) = (-1, 0, 3) + k \cdot (2, -1, 4)$ for $k \in \mathbb{R}$.

$$\text{Parametric equations: } \begin{cases} x = -1 + 2k \\ y = -k \\ z = 3 + 4k \end{cases} \text{ for } k \in \mathbb{R}.$$

$$\text{Continuous equation: } \frac{x+1}{2} = \frac{y}{-1} = \frac{z-3}{4}$$

- b) So that Q belongs to r , we need its coordinates to satisfy the equation of the line, i.e.:

$$\frac{x+1}{2} = \frac{y}{-1} = \frac{(-3)-3}{4},$$

which means:

$$x+1 = -3 \Rightarrow x = -4$$

$$y = \frac{3}{2}$$

Therefore, the coordinates of the point Q are $(-4, \frac{3}{2}, -3)$.

14.

- a) General equation: $\begin{vmatrix} x - (-2) & y - 1 & z - 1 \\ 3 & -1 & 2 \\ 2 & -2 & 1 \end{vmatrix} = 0$. Therefore, the general equation of the plane is $3x + y - 4z + 9 = 0$.

- b) So that Q belongs to the plane, we need its coordinates to verify the equation of the plane, i.e.:

$$3 \cdot (x) + (-3) - 4 \cdot (0) + 9 = 3x + 6 = 0 \Rightarrow x = -2.$$

Therefore, the coordinates of the point Q are $(-2, -3, 0)$.

15.

- a) The matrix $\mathbf{M}$ will be (where each row corresponds to the connections from a website to the other websites, ordered A, B, C, D, E in rows and columns):

$$\mathbf{M} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix}$$

- b) Now, the new matrix is:

$$\mathbf{N} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix}$$

- c) It can be proved that $\mathbf{M}^2 = \mathbf{N}$, i.e., $\mathbf{M}^2$ describes the indirect links through a single connection (site). It can also be proved that $\mathbf{M}^3$ describes the indirect links after two connections, $\mathbf{M}^4$ the indirect links after three connections, etc.

- d) We compute

$$\mathbf{M} + \mathbf{M}^2 = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix} \begin{matrix} A \\ B \\ C \\ D \\ E \end{matrix}$$

This matrix represents the links, either direct or indirect through a single link, between the different websites.

16.

- a) The matrix can be the following (observe that it isn't unique since the dimensions can change):

$$\mathbf{M} = \begin{pmatrix} 13 & 14 & 9 \\ 1 & 1 & 1 \\ 15 & 28 & 28 \\ 1 & 4 & 4 \end{pmatrix}$$

Using this matrix isn't difficult to recover the original text. For example, the value 1 is repeated with enough frequency, which means it's probably a vowel. Logically, the longest the text, the easier it will be to detect patrons in the sequence helping to decode it.

- b) We can get a much more sophisticated codification if we multiply the previous matrix by another matrix (the code-decode matrix):

$$\begin{aligned} \mathbf{M} &= \begin{pmatrix} 13 & 14 & 9 \\ 1 & 1 & 1 \\ 15 & 28 & 28 \\ 1 & 4 & 4 \end{pmatrix} && \leftarrow \text{original sequence} \\ \mathbf{C} &= \begin{pmatrix} 5 & -3 & 2 & 1 \\ -8 & -2 & 7 & 3 \\ -7 & -2 & -4 & 6 \\ 3 & -7 & 5 & 10 \end{pmatrix} && \leftarrow \text{code-decode matrix} \\ \mathbf{C} \cdot \mathbf{M} &= \begin{pmatrix} 93 & 127 & 102 \\ 2 & 94 & 134 \\ -147 & -188 & -153 \\ 117 & 215 & 200 \end{pmatrix} && \leftarrow \text{codified sequence} \end{aligned}$$

It is obvious that, unless the code-decode matrix is already known, it won't be as easy as before to find patrons in the sequence that helps understand the original text.

17.

- a) Considering the matrices obtained after writing the sets of vectors A and B in columns,

we compute their determinants:

$$\begin{vmatrix} 1 & -5 & 7 \\ 1 & -1 & 0 \\ -2 & 2 & -5 \end{vmatrix} = -20 \neq 0$$

$$\begin{vmatrix} 0 & 5 & 6 \\ 1 & -7 & 3 \\ -2 & 4 & 5 \end{vmatrix} = -115 \neq 0$$

Therefore, both sets of vectors form a basis of $\mathbb{R}^3$. In order to write the vector $\mathbf{v} = (2, 1, -1)$ in the basis formed by the set A , we compute:

$$\begin{pmatrix} 1 & -5 & 7 \\ 1 & -1 & 0 \\ -2 & 2 & -5 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ \frac{-3}{5} \\ \frac{-1}{5} \end{pmatrix}$$

Also, to write the vector $\mathbf{v}$ in the basis formed by the set B , we compute:

$$\begin{pmatrix} 0 & 5 & 6 \\ 1 & -7 & 3 \\ -2 & 4 & 5 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{152}{115} \\ \frac{16}{115} \\ \frac{5}{23} \end{pmatrix}$$

b) The change-of-basis matrix from B to A can be computed as:

$$\begin{pmatrix} 1 & -5 & 7 \\ 1 & -1 & 0 \\ -2 & 2 & -5 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 0 & 5 & 6 \\ 1 & -7 & 3 \\ -2 & 4 & 5 \end{pmatrix} = \begin{pmatrix} \frac{5}{4} & -\frac{13}{2} & -\frac{8}{5} \\ \frac{1}{4} & \frac{1}{2} & -\frac{23}{5} \\ 0 & 2 & -\frac{11}{5} \end{pmatrix}$$

We can check how this matrix changes the components of the vector $\mathbf{v}$ from the basis B into the basis A , as we have seen before:

$$\begin{pmatrix} \frac{5}{4} & -\frac{13}{2} & -\frac{8}{5} \\ \frac{1}{4} & \frac{1}{2} & -\frac{23}{5} \\ 0 & 2 & -\frac{11}{5} \end{pmatrix} \cdot \begin{pmatrix} \frac{152}{115} \\ \frac{16}{115} \\ \frac{5}{23} \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ \frac{-3}{5} \\ \frac{-1}{5} \end{pmatrix}$$

18.

a) The matrix $\mathbf{M}$ is (where each row corresponds to the connections from a website to the other websites, ordered A, B, C, D, E and F in rows and columns):

$$M = \begin{pmatrix} \overset{A}{0} & \overset{B}{0} & \overset{C}{0} & \overset{D}{1} & \overset{E}{0} & \overset{F}{0} \\ \overset{A}{0} & \overset{B}{0} & \overset{C}{1} & \overset{D}{0} & \overset{E}{0} & \overset{F}{0} \\ \overset{A}{1} & \overset{B}{0} & \overset{C}{0} & \overset{D}{1} & \overset{E}{1} & \overset{F}{0} \\ \overset{A}{0} & \overset{B}{0} & \overset{C}{0} & \overset{D}{0} & \overset{E}{0} & \overset{F}{1} \\ \overset{A}{0} & \overset{B}{1} & \overset{C}{0} & \overset{D}{0} & \overset{E}{0} & \overset{F}{0} \\ \overset{A}{0} & \overset{B}{0} & \overset{C}{0} & \overset{D}{0} & \overset{E}{1} & \overset{F}{0} \end{pmatrix}$$

b) As explained in exercise 14, the matrix $\mathbf{N}$ has to be $\mathbf{N} = \mathbf{M} + \mathbf{M}^2 + \mathbf{M}^3$, i.e.:

$$\mathbf{N} = \mathbf{M} + \mathbf{M}^2 + \mathbf{M}^3 = \begin{pmatrix} \overset{A}{0} & \overset{B}{0} & \overset{C}{0} & \overset{D}{1} & \overset{E}{1} & \overset{F}{1} \\ \overset{A}{1} & \overset{B}{1} & \overset{C}{1} & \overset{D}{2} & \overset{E}{1} & \overset{F}{1} \\ \overset{A}{1} & \overset{B}{1} & \overset{C}{1} & \overset{D}{2} & \overset{E}{2} & \overset{F}{2} \\ \overset{A}{0} & \overset{B}{1} & \overset{C}{0} & \overset{D}{0} & \overset{E}{1} & \overset{F}{1} \\ \overset{A}{1} & \overset{B}{1} & \overset{C}{1} & \overset{D}{1} & \overset{E}{1} & \overset{F}{0} \\ \overset{A}{0} & \overset{B}{1} & \overset{C}{1} & \overset{D}{0} & \overset{E}{1} & \overset{F}{0} \end{pmatrix}$$

The elements 0 in this matrix represent that neither there's a direct link, neither indirect with one or two links. The elements 1 represent that there's a single link (either direct or indirect between one or two links). Moreover, the elements 2 represent that the link can be done in two different ways (i.e., one with a connection and the other with two, etc). We should also observe that the original graph presents few direct links, but when we consider the connections until order 3 we see that almost all web pages are at a distance of three mouse clicks at most.

19.

a), b), c) The dimension of the matrix is 10×10 :

$$\mathbf{M} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\mathbf{M}^t = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\det(\mathbf{M}) = -9$$

$$\mathbf{M}^{-1} = \begin{pmatrix} 0 & \frac{2}{3} & 0 & -\frac{2}{3} & \frac{1}{3} & 0 & -\frac{1}{3} & 0 & \frac{1}{3} & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & -\frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} & -\frac{1}{3} & 0 & \frac{1}{3} & 0 & -\frac{1}{3} & 0 \\ -\frac{2}{3} & 0 & \frac{2}{3} & 0 & 0 & -\frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & 0 & -\frac{1}{3} & 0 & 0 & \frac{2}{3} & 0 & -\frac{2}{3} & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & 0 & -\frac{1}{3} & \frac{2}{3} & 0 & \frac{1}{3} & 0 & -\frac{1}{3} & 0 \\ -\frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & \frac{2}{3} & 0 & -\frac{1}{3} \\ 0 & -\frac{1}{3} & 0 & \frac{1}{3} & -\frac{2}{3} & 0 & \frac{2}{3} & 0 & \frac{1}{3} & 0 \\ \frac{1}{3} & 0 & -\frac{1}{3} & 0 & 0 & -\frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & -\frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & -\frac{1}{3} & 0 & \frac{1}{3} & 0 \end{pmatrix}$$

d) We have

$$\mathbf{M} \cdot \mathbf{M}^{-1} = \mathbf{I}_{10}$$

$$\mathbf{M}^{-1} \cdot \mathbf{M} = \mathbf{I}_{10}$$

In general matrix product is **not** commutative, even if this particular case it seems to imply it.

Glossary

associate (or adjoint) of an element a_{ij} of a square matrix M Minor complementary determinant of the element a_{ij} together with the sign $(-1)^{i+j}$ for i, j the row and column occupied by the element.

basis of a vector (sub)space Set of linearly independent vectors such that any vector of the (sub)space is a linear combination of them.

complementary minor Given a square matrix, the complementary minor of an element a_{ij} is the minor formed after deleting the row i and column j .

determinant Given a square matrix, the determinant is a number computed using its elements and such that gives information on the linear dependence or independence of its rows (and columns).

dimension of a vector (sub)space Maximum number of linearly independent vectors in the (sub)space, which is equal to the number of elements in any basis of the (sub)space.

generated subspace Given a set of vectors, the generated subspace is the set of all vectors that can be expressed as a linear combination of the vectors on the set.

generators of a vector (sub)space Any set of vectors such that any vector of the (sub)space can be computed as a linear combination of this set.

inverse matrix That presented by square matrices with determinant different from 0. After multiplying (by left or right side) with the given matrix, it gives the identity matrix.

linear independence of vectors Given a set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ of vectors, it is said to be linearly independent if the equation $k_1\mathbf{v}_1 + \dots + k_k\mathbf{v}_k = 0$ only has $k_1 = \dots = k_k = 0$ as a solution. Otherwise, it is said to be linearly dependent.

matrix Table of numbers ordered in n rows and m columns. This would be a matrix $n \times m$.

minor Given a matrix, it is called minor to any determinant of the square matrix formed with elements of the matrix corresponding to k rows and k columns, deleting the other ones. The rows and columns must not necessarily coincide.

norm of a vector The length of a vector. It is computed as the square root of the scalar product of the vector by itself.

orthogonal basis Basis whose vectors are orthogonal two by two.

orthogonal vectors Two vectors are orthogonal if their scalar product is 0.

orthonormal basis Orthogonal basis such that every vector is unitary.

rank of a set of vectors Maximum number of linearly independent vectors on the set.

unitary vector Vector with norm 1.

vector space over $\mathbb{R}$ Set of elements together with an internal operation called sum and a product by real numbers with certain specific properties.

vector Elements of a vector space. The vectors of $\mathbb{R}^n$ can be written as a matrix of a single row (or column) with n elements.

vector subspace Subset of a vector space with structure of vector space.

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